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Research paper

A conformal LLM-guided random forest–LSTM forecaster for uncertainty-aware retail demand and price-scenario decision support

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ABSTRACT

Accurate retail demand forecasting requires reliable point predictions and calibrated uncertainty to support inventory decisions. This study proposes CoRAL-RF, a hybrid framework combining a Random Forest anchor, an LSTM temporal correction expert, frozen Qwen-derived contextual embeddings, a validation-selected correction gate, and conformal prediction intervals. The framework was broadly benchmarked on the M5 dataset and externally validated using anonymized daily unit-demand data from a local retail company, referred to as Company A. On M5, CoRAL-RF achieved MAE = 8.603 and RMSE = 12.506, improving over Random Forest by 2.37% and 1.96%, respectively. On Company A, it achieved MAE = 655.234 and RMSE = 1015.482, corresponding to improvements of 3.62% and 3.24%. Empirical 90% interval coverage reached 0.923 on M5 and 0.919 on Company A. Ablation results showed that Qwen-derived embeddings provided a small, dataset-dependent contribution and mainly supported contextual routing rather than direct forecasting. Inventory simulations indicated lower proxy costs than Random Forest on both datasets under the stated operational assumptions. Overall, CoRAL-RF provides a selective and uncertainty-aware forecasting framework for retail inventory decision support, although its benefits depend on the dataset, evaluation metric, and inventory-policy assumptions.

1. Introduction

1.1. Background and motivation

Retail demand forecasting is a central decision-support problem in supply-chain planning and data-driven retail operations. Forecasts influence replenishment quantities, inventory positioning, safety-stock levels, and service-level decisions. Under-forecasting can generate stockouts, lost sales, emergency replenishment, and reduced customer service, whereas over-forecasting can increase holding costs, obsolescence, waste, and working-capital exposure [1,2]. Consequently, operational forecasting systems should provide not only accurate point predictions but also reliable information about forecast uncertainty that can support inventory decisions.

Retail demand is difficult to forecast because it is influenced by interacting temporal, commercial, and cross-sectional factors. Demand may be intermittent, seasonal, and volatile, while also varying with price, promotions, calendar events, product characteristics, and store-level heterogeneity. Classical models such as ARIMA and SARIMA provide interpretable representations of trend and seasonality but may have difficulty capturing nonlinear interactions and heterogeneous demand regimes [3]. Tree-based models such as Random Forest, XGBoost, and LightGBM are effective for structured retail data and nonlinear covariate relationships [4–6]. Deep-learning and dedicated time-series models, including LSTM, N-BEATS, N-HiTS, PatchTST, and Temporal Fusion Transformer, provide mechanisms for learning sequential and multi-horizon dependencies [7–11]. However, many hybrid approaches combine independently generated forecasts through averaging, stacking, or residual correction without explicitly determining whether a temporal model should modify a strong tabular prediction for a particular observation.

Uncertainty quantification represents an additional challenge. Retail forecasting studies often emphasize point accuracy, although inventory decisions depend on the range of plausible demand outcomes. A model with low average error may still lead to inappropriate replenishment decisions when forecast uncertainty is inadequately represented. Model-agnostic conformal prediction provides a practical method for converting point forecasts into calibrated prediction intervals without imposing a specific probability distribution. Because conformal calibration can be applied to different base forecasters, its contribution should be evaluated separately from that of the underlying forecasting architecture.

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Price and promotion variables further increase the managerial relevance of demand forecasting. These variables are associated with demand variation, inventory depletion, promotion planning, and revenue exposure. Existing research has examined retail price forecasting and machine-learning-assisted pricing decisions using price microdata and commercial covariates [12–14]. The present study does not estimate causal price elasticity or optimize prices. Instead, observed price, promotion, and interaction variables are used as demand predictors, while alternative price inputs are examined only through non-causal sensitivity scenarios.

1.2. Research gap and study positioning

Three research gaps motivate this study.

First, existing hybrid forecasting models frequently combine tabular and temporal predictions through unconditional averaging, stacking, or residual correction. These approaches do not always provide an observation-specific mechanism for controlling when a temporal expert should modify a strong tabular forecast. A selective correction mechanism may be more appropriate when the tabular model is sufficient for most observations and temporal correction is beneficial only under particular demand conditions.

Second, the incremental value of large-language-model representations in structured retail forecasting remains uncertain. Recent studies have explored textual serialization and language-model-based time-series prediction [15,16]. However, directly using an LLM as a numerical forecaster can introduce computational cost, prompt sensitivity, and limited numerical reliability. A more conservative alternative is to use a frozen open-weight language model as an auxiliary contextual encoder. Nevertheless, its contribution must be tested against engineered numerical features, shuffled contextual representations, and conventional categorical embeddings.

Third, evidence obtained from a public benchmark may not transfer directly to a real operational retail environment. Differences in product assortment, demand scale, promotion mechanisms, store organization, data quality, and commercial practices can alter model rankings and the usefulness of contextual routing. Evaluation should therefore combine broad benchmarking on a public dataset with external validation using real company data. Statistical comparisons should also account for dependence among repeated observations from the same products, stores, or forecasting entities.

Accordingly, this study addresses the following research questions:

1. Can a selectively gated LSTM correction improve a strong Random Forest anchor across a public retail benchmark and a real company dataset?
2. Do frozen Qwen-derived contextual embeddings provide incremental routing information beyond no-Qwen, shuffled-Qwen, and conventional categorical-embedding alternatives?
3. Can model-agnostic conformal calibration provide reliable uncertainty estimates and support inventory decisions under explicitly stated operational assumptions?

1.3. Proposed framework and contributions

To address these questions, this paper proposes **CoRAL-RF**, a **Co**nformal **R**outing and **A**daptive **L**LM-guided **R**andom-**F**orest framework for uncertainty-aware retail demand forecasting and inventory decision support.

The framework contains five components. First, a Random Forest produces a tabular forecast from lagged unit demand, rolling statistics, commercial variables, calendar information, and entity-level features. Second, an LSTM estimates a possible temporal deviation from the Random Forest prediction rather than replacing it. Third, a frozen Qwen encoder transforms a serialized retail record into an auxiliary contextual embedding. Fourth, a validation-selected correction gate determines how much of the LSTM deviation should be applied. Finally, a separate conformal-calibration layer converts the selected point forecast into a 90% prediction interval.

The Random Forest therefore remains the principal forecasting anchor, while the LSTM provides a potential temporal correction. Qwen does not directly generate the final numerical forecast; its embedding is used only as an optional input to the correction gate. This structure allows the model to preserve the Random Forest prediction when temporal correction is unnecessary and to apply the LSTM deviation when the gate identifies potentially useful temporal information.

The framework is evaluated in two complementary settings. The M5 dataset is used as the principal development benchmark because it provides detailed daily item–store unit demand, price, calendar, event, and hierarchical information. External validation is conducted using anonymized daily SKU-level unit-demand data obtained from a local retail company, hereafter referred to as *Company A*. Company, product, and store identifiers are replaced with artificial codes to preserve commercial confidentiality.

The study makes the following contributions:

1. It develops an observation-specific Random-Forest–LSTM architecture in which an LSTM correction is selectively applied to a tabular forecasting anchor rather than combined through unconditional averaging.
2. It evaluates frozen Qwen-derived embeddings as auxiliary routing information using no-Qwen, shuffled-Qwen, categorical-embedding, standalone-Qwen, and gate-diagnostic comparisons.
3. It conducts broad benchmarking on the M5 dataset using classical, machine-learning, deep-learning, probabilistic, advanced time-series, and foundation-model baselines. Transferability is subsequently evaluated using anonymized observed unit-demand data from Company A and a focused set of strong tabular, temporal, and contextual controls. The company-data evaluation additionally uses entity-level dependence-aware comparisons.
4. It applies conformal prediction as a distinct model-agnostic calibration stage and evaluates empirical coverage, interval width, and probabilistic or interval scores. This separation avoids attributing the properties of conformal calibration solely to the proposed point-forecasting architecture.
5. It connects forecasts and prediction intervals from both datasets to an illustrative inventory policy and evaluates inventory proxy cost, overstock units, stockout exposure, and fill rate under explicitly stated holding-cost, stockout-penalty, lead-time, and inventory-carryover assumptions. Price-related results are restricted to non-causal sensitivity analysis where price information is available.

The study does not claim that Qwen-derived embeddings uniformly outperform conventional representations or that CoRAL-RF is superior under every metric and dataset. Instead, it examines whether selective integration provides incremental forecasting value, whether contextual information improves correction routing, whether the findings transfer from a public benchmark to a real retail environment, and whether calibrated uncertainty supports inventory decisions under clearly defined assumptions.

1.4. Paper organization

The remainder of the paper is organized as follows. Section 2 reviews retail demand forecasting, hybrid and LLM-assisted forecasting, uncertainty quantification, inventory decision support, and retail price analysis. Section 3 describes the M5 and Company A datasets, preprocessing, feature engineering, CoRAL-RF architecture, control variants, conformal calibration, benchmark models, and evaluation protocol. Section 4 presents point-forecasting performance, uncertainty results, dependence-aware statistical comparisons, ablation and gate diagnostics, and inventory decision analyses. Section 5 interprets the findings, practical implications, and limitations. Section 6 concludes the paper.

2. Literature review

Retail demand forecasting has progressed from classical statistical methods toward machine-learning, deep-learning, probabilistic, and language-model-assisted approaches. This development reflects the complexity of retail demand, which may be affected by seasonality, promotions, price variation, product heterogeneity, calendar effects, store-level differences, and intermittent purchasing behavior. This section reviews the principal forecasting paradigms relevant to CoRAL-RF and examines uncertainty quantification, contextual encoding, price and promotion effects, inventory decision support, and validation using public and operational retail data.

2.1. Classical, machine-learning, and hybrid retail forecasting

Classical time-series models, including ARIMA, SARIMA, exponential smoothing, and Prophet, remain widely used in retail forecasting because of their interpretability, modest computational cost, and capacity to represent trend and seasonal structures [3,17–20]. These approaches are particularly useful when demand patterns are relatively stable and sufficient historical observations are available. Their predictive capacity may nevertheless be limited when demand is nonlinear, promotions produce abrupt changes, or behavior differs substantially across products and stores [21,22]. Classical point-forecasting models also generally require additional procedures when calibrated uncertainty estimates are needed.

Machine-learning methods such as Random Forest, XGBoost, LightGBM, CatBoost, and support vector regression provide greater flexibility for structured retail data [4,23–27]. Tree-based ensembles can represent nonlinear relationships, feature interactions, heterogeneous product behavior, and sparse demand regimes. Their effectiveness often depends on the careful construction of lagged demand, rolling statistics, calendar indicators, price variables, and promotion features [26,28]. However, these models generally have no explicit temporal memory beyond the historical information included through engineered features.

Hybrid forecasting models combine complementary statistical, machine-learning, and deep-learning components. Examples include ARIMA–LSTM, Prophet–LightGBM, ARIMA–Random Forest, and gradient-boosting–neural-network combinations [25,29–31]. Such models can allow one component to capture dominant temporal structures while another represents nonlinear or covariate-driven effects.

Many hybrid approaches, however, combine forecasts through unconditional averaging, stacking, or post-hoc residual correction. These mechanisms do not always determine at the observation level whether a secondary temporal expert should modify a strong tabular forecast. This limitation motivates selective anchor-correction approaches in which a stable tabular model provides the principal prediction and a temporal component contributes only when an adaptive routing mechanism identifies a potentially useful correction.

2.2. Deep-learning and advanced time-series forecasting

Deep-learning models learn nonlinear temporal dependencies directly from sequential data. Recurrent neural networks, particularly LSTM and GRU architectures, have been widely applied to retail demand forecasting because they can represent sequential patterns and lagged effects [7,32–34]. Probabilistic recurrent models such as DeepAR extend this approach by learning predictive distributions rather than only point forecasts [35,36]. N-BEATS and N-HiTS provide alternative mechanisms based on basis expansion and hierarchical interpolation [8,9].

Transformer-based forecasting models, including Temporal Fusion Transformer, Informer, Autoformer, FEDformer, PatchTST, and TimesNet, are designed to capture long-range dependencies, multi-horizon relationships, and interactions among static and dynamic covariates [11,37–40]. These capabilities are relevant in retail environments where demand depends on temporal regularities, price changes, promotions, and holiday effects.

Advanced time-series models may nevertheless require large datasets, careful hyperparameter tuning, substantial computational resources, and separate calibration procedures to produce reliable prediction intervals. Their relative performance can also change according to the forecasting horizon, aggregation level, dataset structure, and evaluation metric.

The present study therefore does not position CoRAL-RF as a universal replacement for advanced time-series models. Broad comparisons with dedicated time-series alternatives are conducted on the M5 benchmark, while transferability is evaluated separately using anonymized observed unit-demand data from a local retail company, referred to as Company A, and a focused set of strong control models.

2.3. Probabilistic forecasting and uncertainty quantification

Point forecasts alone may be insufficient for inventory decisions because managers must account for the range of plausible demand outcomes. Forecast errors can generate excess inventory, stockouts, additional holding costs, lost sales, and reduced service levels [1,41]. Probabilistic forecasting and uncertainty quantification are therefore central to risk-aware retail planning.

Bayesian Neural Networks represent model parameters probabilistically and produce posterior predictive distributions rather than deterministic outputs [42–44]. Monte Carlo dropout, Bayesian inference, and ensemble-based methods have also been used to approximate predictive uncertainty in neural forecasting models [43,45]. Quantile regression and probabilistic deep-learning methods provide further mechanisms for estimating asymmetric demand ranges.

Conformal prediction offers a model-agnostic alternative that can construct prediction intervals around different base forecasters without requiring a fully specified demand distribution [46,47]. This property is relevant in retail contexts where residual behavior may differ across products, stores, promotions, holidays, and demand regimes.

Empirical coverage should be evaluated jointly with interval width. High coverage may be obtained through excessively wide intervals, whereas narrow intervals may fail to contain realized demand sufficiently often. Appropriate evaluation should therefore consider empirical coverage, average interval width, and probabilistic or interval scores.

Because conformal prediction can be applied to different forecasting models, near-nominal coverage primarily demonstrates the effectiveness of the calibration procedure rather than the superiority of a particular point-forecasting architecture. The present study consequently treats conformal prediction as a separate calibration stage and distinguishes point-forecast accuracy from interval quality.

2.4. LLM-guided and open-weight contextual forecasting

Large language models have recently been explored for time-series forecasting through textual serialization, prompt-based inference, reprogramming, and alignment between numerical and textual representations [15,16,48–50]. These approaches investigate whether pretrained language representations can contribute when numerical observations and contextual variables are converted into structured text.

Using an LLM as a standalone numerical forecaster remains uncertain because language models are primarily pretrained on textual corpora rather than numerical temporal processes. Direct LLM forecasting may also introduce computational cost, prompt sensitivity, serialization dependence, and limited numerical reliability.

A more conservative strategy is to use a frozen open-weight language model as an auxiliary contextual encoder. A serialized retail observation can combine product identity, store information, recent demand, calendar conditions, price, and promotion status where these variables are available. The encoder then transforms this representation into a fixed embedding without directly generating the final numerical forecast.

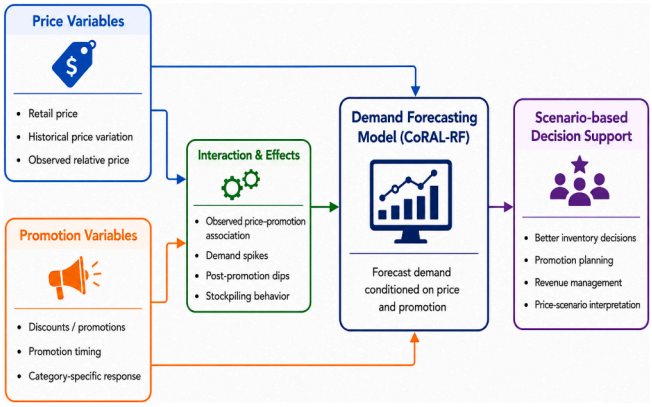


Fig. 1. Role of observed price and promotion variables in the CoRAL-RF forecasting and decision-support pipeline. Price and promotion variables, where available, condition the demand forecast together with temporal, calendar, product, and store information. The resulting forecast and calibrated interval support inventory evaluation and non-causal sensitivity analysis.

However, the same information can also be represented through engineered numerical variables, one-hot encodings, or trainable categorical embeddings. The usefulness of an LLM-derived representation should therefore be established through controlled comparisons rather than assumed.

Relevant controls include removing the contextual representation, shuffling embeddings across observations, and replacing the LLM-derived representation with conventional categorical embeddings. These comparisons indicate whether any improvement depends on contextual information, correct observation–context alignment, or the pretrained language representation itself. A standalone Qwen baseline may additionally be used to determine whether Qwen is competitive as an independent forecasting representation.

In CoRAL-RF, frozen Qwen-derived embeddings are used only as auxiliary inputs to the correction gate. Random Forest and LSTM remain responsible for the tabular anchor and temporal correction, respectively. The Qwen component is therefore evaluated as an incremental routing mechanism rather than as the principal forecasting model.

2.5. Retail price, promotion, and demand interaction

Retail demand is closely associated with price and promotion conditions. Prices affect revenue, purchasing behavior, inventory depletion, and replenishment exposure, while promotions may change both demand timing and volume. Retail analytics studies have examined price forecasting, promotion-response modeling, web-scraped price microdata, causal pricing, and machine-learning-assisted pricing decisions [12–14,51,52].

Temporary discounts may generate demand spikes, post-promotion reductions, stockpiling behavior, and category-specific responses [53–55]. In demand-forecasting models, observed price and promotion variables can therefore be combined with calendar indicators, interaction terms, and lagged demand to condition the forecast on recorded commercial circumstances.

Fig. 1 illustrates this role in the proposed framework. Price and promotion variables are used where available in each dataset. Their inclusion supports demand forecasting and scenario-based managerial interpretation, but it does not establish causal price elasticity.

Changing a historical price input while holding the remaining predictors fixed produces a model-based sensitivity scenario rather than an identified causal response. Causal price-response estimation would require an appropriate identification strategy, such as randomized price variation, instrumental variables, or another causal-inference design. The present study therefore restricts price-related interpretation to non-causal sensitivity analysis.

2.6. External validation and generalizability in retail forecasting

Generalizability is an important concern in retail forecasting because model performance can vary with demand scale, product assortment, aggregation level, promotion mechanisms, data quality, and operational practices. Consequently, a model that performs strongly on a public benchmark may not maintain the same performance in a real retail environment [18,56].

External validation using operational company data complements benchmark evaluation by examining whether a forecasting architecture transfers to demand patterns and commercial conditions not represented during model development. However, validation using a single company dataset should be interpreted as evidence of transferability to one additional setting rather than proof of universal applicability [56].

Forecast evaluation must also account for the temporal and cross-sectional dependence of retail observations. Daily errors from the same product, store, or product-store series are generally not independent, and inappropriate data partitioning or observation-level statistical testing may overstate model performance. Chronological holdout evaluation and dependence-aware comparisons based on forecasting entities therefore provide a more reliable assessment of model generalizability [57,58].

2.7. Inventory decision support

Forecasting performance becomes operationally meaningful when predictions are translated into inventory decisions. Unit-demand forecasts, demand quantiles, or upper prediction bounds can be used to define replenishment or coverage quantities and evaluate holding cost, shortage cost, overstock units, stockout units, and fill rate [1,41].

Because both M5 and Company A use recorded unit-demand outcomes as the target, inventory indicators can be interpreted consistently across the two datasets. These observations may represent sales or orders rather than unconstrained latent demand, particularly when stockouts are not explicitly corrected. Inventory outcomes also remain dependent on the selected decision policy and assumptions. Holding and shortage cost ratios, replenishment lead time, inventory carryover, service-level targets, backlogging, lost-sales treatment, capacity restrictions, and minimum-order quantities may alter model rankings.

Inventory simulations should therefore be interpreted as scenario-specific decision-support evidence rather than universal operational savings. The present study evaluates an illustrative upper-interval inventory policy under explicitly stated assumptions and compares CoRAL-RF with alternative conformalized forecasters where equivalent outputs are available.

2.8. Research gap and positioning of CoRAL-RF

The literature identifies five gaps addressed in this study:

1. Hybrid forecasting approaches commonly use averaging, stacking, or residual correction, while observation-specific mechanisms for selectively modifying a strong tabular anchor remain less established.
2. The incremental value of frozen LLM-derived representations in structured retail forecasting remains uncertain and requires comparison with no-context, shuffled-context, and conventional categorical alternatives.
3. Uncertainty evaluation does not always distinguish the quality of the point forecaster from the contribution of the calibration procedure.
4. Forecasting architectures are frequently evaluated on a single public dataset without operational external validation or dependence-aware statistical comparison.
5. Forecasting performance is not always connected to inventory outcomes under transparent and explicitly stated decision assumptions.

CoRAL-RF addresses these gaps by combining a Random Forest anchor, an LSTM temporal correction expert, an optional frozen Qwen-

Table 1
Positioning of CoRAL-RF relative to representative retail forecasting paradigms.

Forecasting paradigm	Representative methods	Main strength	Uncertainty handling	Price/promotion handling	Main limitation or evaluation requirement	Key references
Classical statistical models	ARIMA, SARIMA, ETS, Prophet	Interpretable trend and seasonality modeling	Limited or model-specific	Usually included as exogenous regressors	Limited flexibility for nonlinear interactions and heterogeneous SKU- or store-level behavior	[3,17–20]
Machine-learning models	RF, XGBoost, LightGBM, CatBoost, SVR	Strong nonlinear learning from structured tabular data	Requires quantile, ensemble, Bayesian, or conformal extensions	Effective with engineered price and promotion features	Limited explicit temporal memory; performance depends on feature engineering	[4,23–28]
Deep-learning models	LSTM, GRU, DeepAR, CNN–LSTM, N-BEATS, N-HiTS	Temporal-dependency learning and flexible nonlinear modeling	Available in probabilistic variants or through external calibration	Requires explicit representation of commercial covariates	Computational and tuning requirements; performance may vary across datasets and metrics	[7–9,32–36]
Transformer time-series models	TFT, Informer, Autoformer, FEDformer, PatchTST, TimesNet	Long-range dependency and multi-horizon modeling	Usually through quantile or probabilistic heads	Can incorporate dynamic covariates	Large training requirements and no guaranteed superiority on structured retail data	[11,37–40]
LLM-assisted forecasting	LLM4TS, Time-LLM, zero-shot serialization, LLMForecaster	Textual serialization and pretrained contextual representation	Generally indirect or requires external calibration	Can encode commercial, calendar, event, and entity context	Computational cost, numerical reliability, and uncertain incremental benefit beyond categorical or tabular representations	[15,16,48–50,59]
Probabilistic and conformal forecasting	BNN, MC dropout, quantile regression, conformal prediction, CQR	Prediction intervals and risk-aware decision support	Central objective	Model-agnostic and applicable to demand conditioned on observed commercial variables	Calibration is separable from base-model accuracy; coverage should be assessed jointly with interval width and proper scores	[41–47]
Retail price and promotion analytics	Price forecasting, promotion response, causal pricing, optimization models	Commercial-driver, revenue, and promotion analysis	Usually secondary to pricing or optimization objectives	Price and promotion have a central role	Predictive price scenarios do not establish causal elasticity or direct price optimality	[12–14,51–55]
Proposed CoRAL-RF	**RF anchor + LSTM correction + frozen Qwen routing input + conformal calibration**	**Selective integration of tabular, temporal, contextual, and uncertainty components**	**Separate model-agnostic conformal layer**	**Price, promotion, calendar, and interaction features where available**	**Incremental value assessed through controls, gate diagnostics, M5 benchmarking, Company A external validation, and inventory simulations**	[4,7,46,47,59]

derived routing representation, and a separate conformal-calibration layer. The framework is broadly benchmarked on M5 and externally validated using anonymized observed unit-demand data from Company A. Its incremental components are assessed through controlled ablations, contextual alternatives, gate diagnostics, matched calibration comparisons where available, entity-level statistical analyses, and inventory simulations under stated assumptions.

Table 1 summarizes the position of CoRAL-RF relative to the principal forecasting paradigms discussed in the literature.

3. Methodology

This section presents the experimental design and the proposed CoRAL-RF framework for uncertainty-aware retail demand forecasting. The methodology combines: (i) model development and broad benchmarking on the M5 retail forecasting dataset [60,61]; (ii) external validation using anonymized daily SKU-level unit-demand data from Company A; (iii) a Random Forest–LSTM anchor-correction architecture in which frozen Qwen-derived embeddings are evaluated as auxiliary routing information; (iv) model-agnostic conformal calibration; and (v) diagnostic, computational, and inventory-oriented evaluation. The complete workflow is summarized in Fig. 2.

3.1. Experimental workflow

The experimental workflow contains five stages. First, each dataset is transformed into a leakage-safe daily panel ordered chronologically by forecasting entity and date. M5 combines item-store sales, calendar records, sell-price information, and hierarchical metadata, whereas Company A provides anonymized daily SKU-level unit demand and the associated operational predictors retained in the experiment data.

Second, dataset-specific numerical features are constructed from lagged sales, rolling statistics, calendar indicators, commercial variables, and entity metadata. In parallel, the same information is serialized into structured text and passed through a frozen Qwen encoder. The resulting Qwen representation is used only as an auxiliary input to the correction gate and does not directly generate the final demand forecast.

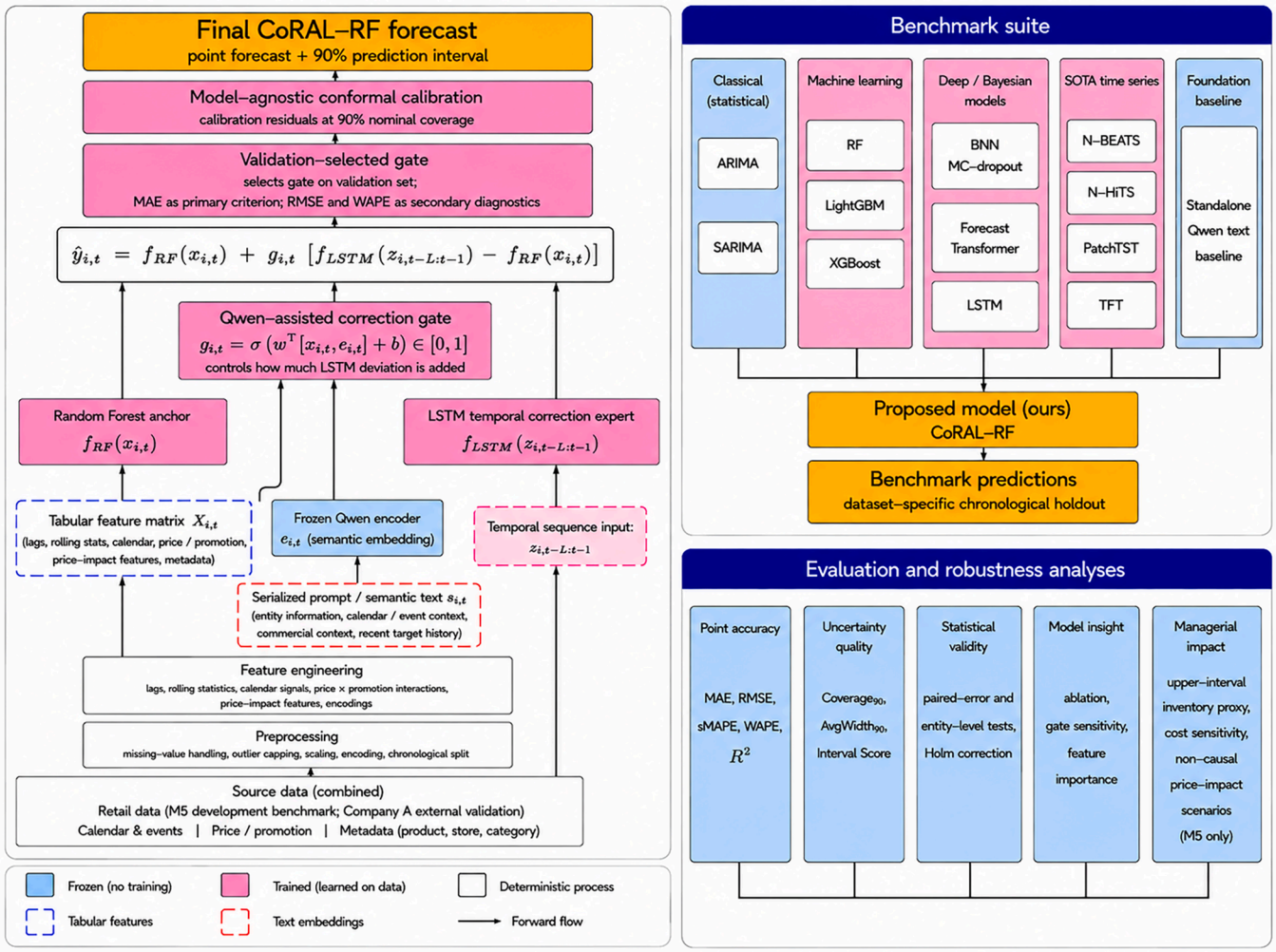


Fig. 2. Two-dataset experimental workflow of CoRAL-RF. M5 is used as the principal development and broad-benchmark dataset, while anonymized Company A data provide external validation. The pipeline includes chronological data preparation, numerical feature engineering, contextual serialization, benchmark and control models, the RF LSTM correction architecture, model-agnostic conformal calibration, gate diagnostics, computational evaluation, and inventory decision analysis.

Third, the Random Forest anchor, LSTM correction expert, CoRAL-RF model, benchmark models, and LLM-control variants are trained under chronological training–validation/calibration–test protocols. Model and gate configurations are selected using the middle partition only, without access to the held-out test outcomes.

Fourth, conformal calibration is applied after point forecasting. On Company A, the same calibration procedure and nominal coverage level are also applied to strong non-LLM baselines, enabling the contribution of conformal calibration to be distinguished from that of the hybrid point forecaster.

Fifth, held-out forecasts are evaluated using point accuracy, uncertainty quality, component ablations, gate diagnostics, regime analysis, computational cost, and inventory indicators [1,62,63]. Statistical claims are restricted to analyses for which the required dependence-aware paired errors are available.

3.2. Datasets and computational environment

3.2.1. M5 development benchmark

The M5 retail forecasting dataset is used as the principal model-development and broad-benchmark environment. It contains daily item store unit sales, calendar information, sell-price records, and hierarchical product and store metadata [60,61]. The complete benchmark contains 30,490 item store demand series corresponding to 3049 products sold across 10 stores in three U.S. states. The sales history contains 1941 daily observations, the calendar contains 1969 daily records, and the sell-price table contains more than 6.8 million item store week records [60,61].

M5 is used to compare CoRAL-RF with a broad suite of classical, machine-learning, deep-learning, probabilistic, advanced time-series, and foundation-model baselines. The chronological observations are divided into 60% training, 20% validation/calibration, and 20% held-out testing. The middle partition is used for configuration selection and conformal calibration; consequently, the reported interval coverage is treated as empirical held-out performance rather than as a strict finite-sample coverage guarantee based on a calibration set independent of model selection. No future observation is used to construct lagged or rolling features for an earlier prediction date.

Table 2 summarizes the M5 development dataset and experimental setting.

3.2.2. Company A external validation

An anonymized operational dataset supplied by a local retailer, referred to as Company A, is used to evaluate whether the proposed architecture transfers beyond the public M5 benchmark. The target is observed daily unit demand at SKU level. Because the retained data do not explicitly correct observations for stockouts, the target is not interpreted as unconstrained latent demand. Company, SKU, and location identifiers

Table 2
Summary of the M5 development dataset.

Item	Description
Dataset	M5 Forecasting Accuracy retail dataset [60,61]
Forecasting unit	Daily item store demand series
Demand series	30,490 item store series [61]
Products	3,049 items [61]
Stores and regions	10 stores in three U.S. states [60,61]
Sales horizon	1,941 daily sales observations [61]
Calendar data	Weekday, month, events, holidays, and SNAP indicators [60]
Commercial data	More than 6.8 million item store week sell-price observations [60,61]
Target	Daily unit sales
Principal covariates	Lagged sales, rolling statistics, sell price, event and calendar indicators, category, department, store, and state
Experimental role	Model development, broad baseline comparison, ablation, and M5 decision analysis
Split	Chronological 60/20/20 training–validation/calibration–test split

Table 3
Summary of the Company A external-validation setting.

Item	Description
Dataset	Anonymized operational data from Company A
Forecasting unit	Daily SKU-level unit demand
Validation scope	Fixed anonymized forecasting entities
Held-out observations	17,793 daily observations
Inferential unit	Forecasting entity
Target	Observed daily unit demand
Principal covariates	Lagged demand, rolling statistics, calendar variables, and the anonymized operational predictors retained in the experiment data
Experimental role	External validation, LLM controls, matched conformal comparison, gate diagnostics, and inventory proxy analysis
Entity selection	Fixed external-validation sample of 150 anonymized forecasting entities available in the retained experiment outputs
Observation period	Chronologically ordered daily observations; calendar dates are withheld for commercial confidentiality
Split	Chronological training, validation/calibration, and held-out test partitions without random shuffling; calendar boundaries are not disclosed

are replaced with artificial codes to preserve commercial confidentiality.

All competing models and control variants use the same forecasting entities and chronological train, validation/calibration, and held-out test boundaries. The held-out evaluation contains 17,793 daily observations. Observations are ordered chronologically within each forecasting entity, and no future value is used to construct an earlier predictor.

Company A is treated as external validation rather than as an additional broad benchmark. Its purpose is to test transferability, LLM-control behavior, gate selectivity, matched conformal calibration, entity-level performance, and inventory decision indicators. Because the retained Company A experiment does not contain the item-level sell-price structure used in the M5 price scenarios, price-scenario analysis is performed only on M5.

Table 3 summarizes the Company A external-validation setting.

3.2.3. Computational environment

All experiments were implemented in Python using Google Colab with an NVIDIA Tesla T4 GPU [64]. Data preparation and numerical computation used NumPy and pandas; statistical and machine-learning models used scikit-learn, statsmodels, XGBoost, and LightGBM; neural and contextual models used PyTorch and Hugging Face Transformers; and quantitative figures were generated using matplotlib [5,6,65–71].

Table 4 summarizes the computational environment and reproducibility controls.

Table 4
Computational environment and reproducibility controls.

Component	Specification
Programming language	Python 3.12.13
Platform	Google Colab [64]
GPU	NVIDIA Tesla T4
Data processing	NumPy and pandas [65,66]
Machine learning	scikit-learn, XGBoost, and LightGBM [5,6,67]
Deep learning and LLM	PyTorch and Hugging Face Transformers [69,70]
Statistical modeling	statsmodels [68]
Visualization	matplotlib [71]
Uncertainty method	Model-agnostic conformal prediction [72,73]
M5 validation protocol	Chronological 60/20/20 training–validation/calibration–test split; empirical interval coverage is reported because model selection and calibration use the middle partition
Company A validation protocol	Fixed sample of 150 anonymized forecasting entities evaluated using chronological training, validation/calibration, and held-out test partitions
Reproducibility controls	Fixed random seeds, validation-only configuration selection, common test windows, saved manifests, and frozen Qwen parameters
Computational measures	Training time, inference time, peak GPU memory, and trainable parameter count

3.3. Feature engineering and contextual serialization

For dataset d, forecasting entity i, and date t, the numerical feature vector is denoted by $\mathbf{x}_{i,t}^{(d)}$. Only information observable before or at the forecasting origin is used.

For M5, the lag vector contains the preceding 28 demand observations:

$$\mathbf{L}_{i,t}^{M5} = [y_{i,t-1}, y_{i,t-2}, \ldots, y_{i,t-28}]. \tag{1}$$

Short- and medium-term behavior is summarized using 7- and 28-day rolling statistics:

$$\mathbf{R}_{i,t}^{(w)} = \left[\mu_{i,t}^{(w)}, \sigma_{i,t}^{(w)}\right], \qquad w \in \{7, 28\}, \tag{2}$$

where $\mu_{i,t}^{(w)}$ and $\sigma_{i,t}^{(w)}$ denote the leakage-safe rolling mean and standard deviation calculated from observations preceding date t.

The M5 feature vector additionally includes sell price, promotion or event indicators, SNAP variables, calendar variables, and hierarchical item store metadata. The price promotion interaction is:

$$z_{i,t}^{price\times promo} = p_{i,t}\, promo_{i,t}. \tag{3}$$

The M5 tabular representation is:

$$\mathbf{x}_{i,t}^{M5} = [\mathbf{L}_{i,t}, \mathbf{R}_{i,t}^{(7)}, \mathbf{R}_{i,t}^{(28)}, p_{i,t}, promo_{i,t}, z_{i,t}^{price\times promo}, \mathbf{c}_t, \mathbf{m}_i]. \tag{4}$$

For Company A, the same leakage-safe principle is used, with predictors restricted to the anonymized operational variables retained in the experiment data:

$$\mathbf{x}_{i,t}^{A} = [\mathbf{L}_{i,t}^{A}, \mathbf{R}_{i,t}^{A}, \mathbf{c}_t, \mathbf{m}_i, \mathbf{k}_{i,t}], \tag{5}$$

where $\mathbf{c}_t$ contains calendar variables, $\mathbf{m}_i$ contains anonymized entity metadata, and $\mathbf{k}_{i,t}$ contains the time-varying operational variables available for Company A. Because the company data and its detailed field definitions are commercially restricted, the external-validation description is limited to these predictor families. This restriction limits exact independent reconstruction of the Company A preprocessing pipeline and is acknowledged in Section 5.9.

In parallel, each observation is converted into a structured textual record:

$$s_{i,t}^{(d)} = \text{Serialize}\left(\mathbf{m}_i, \mathbf{c}_t, \mathbf{v}_{i,t}^{(d)}, \mathbf{L}_{i,t}^{(d)}\right), \tag{6}$$

Table 5
Benchmark and control models used in the two experimental settings.

Family	Models	M5 role	Company A role	Evaluation purpose
Classical statistical	ARIMA, SARIMA	Included	Not included in the focused external validation	Interpretable autoregressive and seasonal baselines [3]
Machine learning	Random Forest, XGBoost, LightGBM	RF, XGBoost, and LightGBM included	Random Forest and LightGBM included	Strong nonlinear tabular comparison [4–6]
Deep learning	LSTM, Forecast Transformer	Both included	LSTM included	Sequential and attention-based temporal comparison [7,37]
SOTA time series	N-BEATS, N-HiTS, PatchTST, TFT	Included	Not included in the focused external validation	Modern dedicated time-series comparison [8–11]
Probabilistic	BNN with MC dropout	Included	Not included	Neural uncertainty-aware comparison [43]
Foundation-model baseline	Standalone Qwen text fallback	Included	Not used as the principal Company A control	Tests direct text-to-number forecasting [59]
Context controls	No-Qwen gate, shuffled-Qwen gate, categorical-embedding gate	No-Qwen control included; standalone Qwen also reported	All three gate controls included	Tests whether improvement depends on Qwen, correct context alignment, or a conventional categorical representation
Proposed model	CoRAL-RF	Included	Included	RF anchor, LSTM correction, Qwen-derived routing input, and separate conformal calibration

where $\mathbf{v}_{i,t}^{(d)}$ contains the commercial and event variables available in dataset d. For M5, this includes sell price, promotion or event status, and recent unit demand. For Company A, it includes the anonymized operational variables and recent unit demand available in the retained data.

The serialized record is passed through a frozen Qwen encoder:

$$\mathbf{e}_{i,t} = f_{\text{Qwen}}(s_{i,t}), \tag{7}$$

where no Qwen parameters are updated during forecasting-model training. The embedding is treated as an auxiliary representation whose contribution must be established through controlled comparisons. It is not assumed to be superior to engineered numerical features or conventional categorical embeddings.

3.4. Benchmark models and LLM controls

The M5 experiment uses a broad benchmark suite because it is the principal development setting. The Company A experiment uses a focused group of strong tabular, temporal, and contextual controls to evaluate transferability and the incremental role of Qwen. Models within each dataset use the same chronological partitions and target observations. The same engineered predictors are supplied where compatible with the model family, while sequence-specific and text-specific representations are restricted to models that require them.

Table 5 summarizes the benchmark and control models used in the two experimental settings.

The principal contextual controls are defined as follows.

No-Qwen gate. The Qwen embedding is removed, and the correction gate is estimated using only numerical and engineered tabular variables.

Shuffled-Qwen gate. The Qwen embeddings are randomly reassigned across observations while preserving their marginal distribution:

$$\tilde{\mathbf{e}}_{i,t} = \mathbf{e}_{\pi(i),t}, \tag{8}$$

where $\pi(\cdot)$ is a fixed random permutation. This control tests whether performance depends on the correct correspondence between the contextual representation and the observation.

Categorical-embedding gate. The Qwen representation is replaced with trainable embeddings constructed from conventional store, product, category, calendar, or promotion identifiers:

$$\mathbf{e}_{i,t}^{cat} = f_{cat}(\mathbf{m}_i, \mathbf{c}_t, promo_{i,t}). \tag{9}$$

This comparison tests whether the pretrained language representation contributes beyond a simpler categorical alternative.

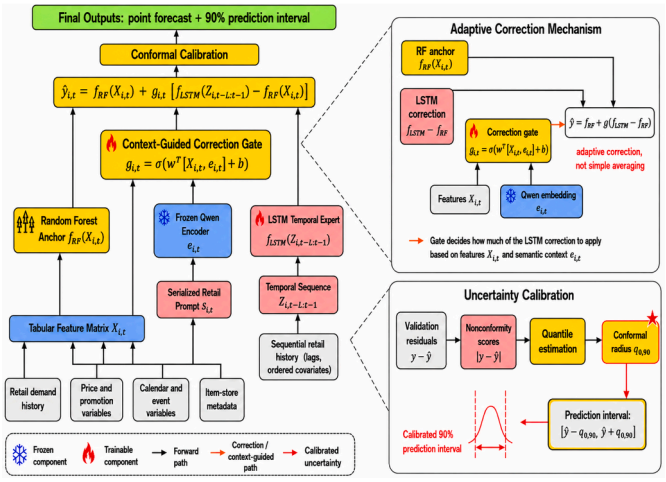


Fig. 3. Point-forecasting architecture and separate uncertainty layer of CoRAL-RF. Random Forest provides the tabular anchor, LSTM estimates a possible temporal correction, and the validation-selected gate uses numerical features and an optional frozen Qwen-derived contextual representation. Conformal calibration is subsequently applied as a separate model-agnostic stage.

Standalone Qwen baseline. On M5, a direct serialized-context Qwen fallback is included to determine whether the language model is competitive as an independent numerical forecaster. This baseline is conceptually different from the proposed use of Qwen as an auxiliary gate input.

3.5. Proposed CoRAL-RF point-forecasting architecture

CoRAL-RF uses a model-level anchor correction structure. The point-forecasting architecture contains a Random Forest anchor, an LSTM temporal expert, a frozen Qwen contextual encoder, and an adaptive correction gate. Conformal prediction is not part of the point-forecasting architecture itself; it is applied afterward as a separate model-agnostic calibration procedure. Fig. 3 illustrates the complete CoRAL-RF point-forecasting architecture and separate uncertainty layer.

Let $y_{i,t}$ denote observed demand or sales and let $\mathbf{x}_{i,t}$ denote the dataset-specific engineered feature vector.

3.5.1. Random forest tabular anchor

The Random Forest anchor is:

$$\hat{y}_{i,t}^{RF} = f_{RF}(\mathbf{x}_{i,t}). \tag{10}$$

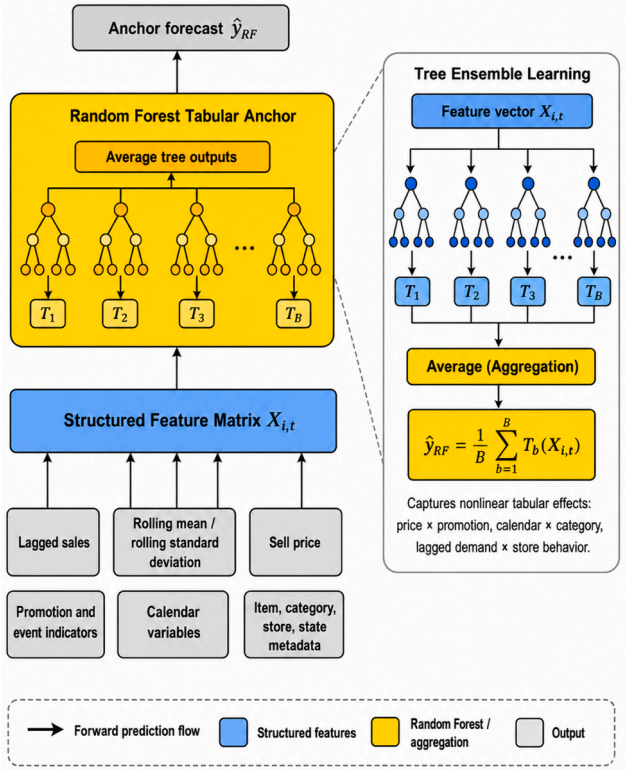


Fig. 4. Random Forest tabular anchor. The model processes lagged demand, rolling statistics, calendar, commercial, and entity-level features to produce the reference forecast.

For a forest containing B regression trees:

$$\hat{y}_{i,t}^{RF} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}_{i,t}), \tag{11}$$

where $T_b(\cdot)$ is the output of tree b. The Random Forest prediction remains the principal forecast unless the correction gate assigns positive weight to the LSTM deviation. Fig. 4 illustrates the Random Forest tabular-anchor module.

3.5.2. LSTM temporal correction expert

Let $\mathbf{z}_{i,t-L:t-1}$ denote the ordered sequence observed over a look-back horizon L. The LSTM output is:

$$\hat{y}_{i,t}^{LSTM} = f_{LSTM}(\mathbf{z}_{i,t-L:t-1}). \tag{12}$$

The temporal correction is defined relative to the Random Forest anchor:

$$\Delta_{i,t}^{LSTM} = \hat{y}_{i,t}^{LSTM} - \hat{y}_{i,t}^{RF}. \tag{13}$$

Thus, the LSTM is not unconditionally averaged with the Random Forest. It proposes a deviation that may be suppressed or partially applied by the gate.

Fig. 5 illustrates the LSTM temporal-correction expert and its relationship to the Random Forest anchor.

3.5.3. Frozen Qwen contextual encoder

The frozen Qwen encoder transforms the serialized record into $\mathbf{e}_{i,t}$ according to Eq. 7. The language model does not directly predict the target, and its parameters are not updated during gate training. This design reduces training cost and improves reproducibility, while the no-Qwen, shuffled-Qwen, and categorical-embedding controls determine whether the representation contributes incremental routing information.

Fig. 6 illustrates the frozen Qwen contextual encoder and its role as an auxiliary gate input.

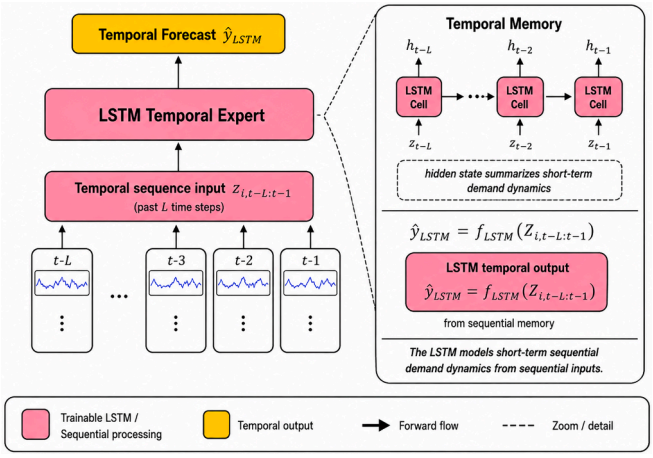


Fig. 5. LSTM temporal correction expert. The module produces a possible deviation from the Random Forest anchor rather than independently determining the final forecast.

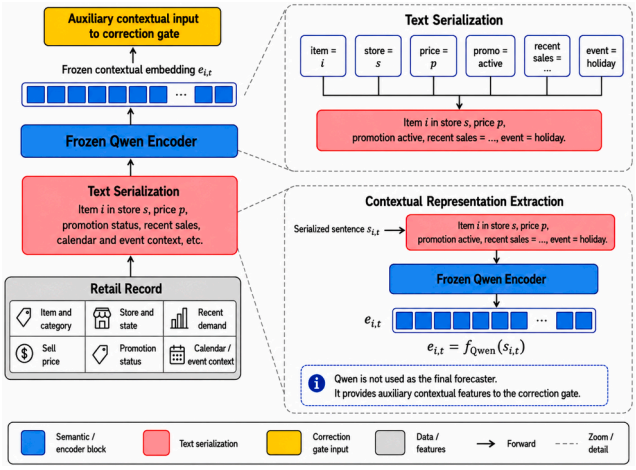


Fig. 6. Frozen Qwen contextual encoder. Structured retail information is serialized and encoded as an auxiliary gate input. The encoder does not directly generate the final demand or sales prediction.

3.5.4. Adaptive correction gate

For the full model, the correction weight is:

$$g_{i,t} = \sigma\left(\mathbf{w}^{\top}[\mathbf{x}_{i,t}, \mathbf{e}_{i,t}] + b\right), \tag{14}$$

where $g_{i,t} \in [0,1]$ and $\sigma(\cdot)$ is the sigmoid function.

The final point forecast is:

$$\hat{y}_{i,t} = \hat{y}_{i,t}^{RF} + g_{i,t}\Delta_{i,t}^{LSTM}. \tag{15}$$

Substituting Eqs. 10 and 13 gives:

$$\hat{y}_{i,t} = f_{RF}(\mathbf{x}_{i,t}) + g_{i,t}\left[f_{LSTM}(\mathbf{z}_{i,t-L:t-1}) - f_{RF}(\mathbf{x}_{i,t})\right]. \tag{16}$$

A gate value close to zero preserves the Random Forest anchor, while a larger value applies more of the LSTM deviation. The architecture permits sparse intervention; frequent activation is not required for the model to be valid. Whether the correction is actually selective and beneficial is evaluated through gate-quartile and regime diagnostics rather than assumed from the architecture.

The Random Forest and LSTM components are fitted on the training partition and then held fixed. Candidate gate configurations are evaluated exclusively on the middle validation/calibration partition, and the selected configuration is fixed before final testing. Conformal residuals are also obtained from this middle partition. Because configuration selection and calibration are not based on independent subsets, the reported coverage is interpreted empirically and no strict finite-sample

guarantee conditional on model selection is claimed. The retained experiment records do not support a reproducible claim about a separately optimized gradient-based gate-training schedule; therefore, optimizer-specific settings are not asserted here.

3.6. Validation-safe configuration selection

All preprocessing transformations, model configurations, gate parameters, and control variants are selected without access to the held-out test outcomes. For M5, the chronological 60/20/20 training–validation/calibration–test split is maintained for all models. For Company A, all models use the same anonymized forecasting entities and chronological boundaries defined in the experiment manifest.

Candidate configurations are ranked using their recorded validation forecasting performance, with MAE as the primary selection criterion and RMSE and WAPE used as secondary diagnostics. No weighted combination of differently scaled metrics is claimed. The gate configuration and all hyperparameters are fixed before final test evaluation. The held-out test periods are used only for reporting the final forecasting, uncertainty, statistical, and decision-support results.

3.7. Model-agnostic conformal calibration

Conformal calibration is applied after the point-forecast model has been fixed. For each observation in the chronological validation/calibration period, the absolute residual is:

$$r_{i,t} = |y_{i,t} - \hat{y}_{i,t}|. \tag{17}$$

Let $n_{cal} = |\mathcal{D}_{cal}|$ and let $r_{(1)} \le \cdots \le r_{(n_{cal})}$ denote the ordered calibration residuals. For nominal coverage $1-\alpha$, define the finite-sample conformal rank and radius as:

$$k = \min\{n_{cal}, \lceil (n_{cal}+1)(1-\alpha) \rceil\}, \qquad q_{1-\alpha} = r_{(k)}. \tag{18}$$

The prediction interval is:

$$\hat{C}_{i,t} = [\hat{L}_{i,t}, \hat{U}_{i,t}] = [\hat{y}_{i,t} - q_{1-\alpha}, \hat{y}_{i,t} + q_{1-\alpha}]. \tag{19}$$

The target coverage is 90%, corresponding to $\alpha = 0.10$. Because conformal prediction is model-agnostic, interval validity is interpreted as a property of the calibration procedure applied to the selected point forecaster rather than as evidence attributable solely to the Random Forest–LSTM–Qwen architecture. Because the retained protocol reuses the middle partition for configuration selection and calibration, the usual split-conformal finite-sample guarantee does not apply without qualification; the manuscript therefore reports empirical coverage on the untouched test period.

On M5, complete calibration metrics are retained only for CoRAL-RF. An archived Random Forest upper-endpoint output is available for the inventory proxy analysis, but its corresponding coverage, width, and interval-score outputs were not retained. It is therefore used only for the stated inventory comparison, and no M5 claim of cross-model calibration or interval superiority is made.

On Company A, the same nominal level, chronological calibration period, residual definition, and interval construction are applied to CoRAL-RF, Random Forest, and LightGBM. The contextual control variants are also evaluated using the corresponding calibrated intervals. This matched design allows point-forecast quality and calibration quality to be examined separately.

3.8. Evaluation metrics

3.8.1. Point-forecast metrics

Point accuracy is evaluated using MAE, RMSE, sMAPE, WAPE, and R^2 [61,62].

$$\mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n} |y_j - \hat{y}_j|. \tag{20}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n} (y_j - \hat{y}_j)^2}. \tag{21}$$

$$\mathrm{WAPE} = \frac{\sum_{j=1}^{n} |y_j - \hat{y}_j|}{\sum_{j=1}^{n} |y_j|} \times 100. \tag{22}$$

$$\mathrm{sMAPE} = \frac{100}{n}\sum_{j=1}^{n} \frac{2|y_j - \hat{y}_j|}{|y_j| + |\hat{y}_j| + \epsilon}, \tag{23}$$

where ϵ prevents division by zero.

$$R^2 = 1 - \frac{\sum_{j=1}^{n} (y_j - \hat{y}_j)^2}{\sum_{j=1}^{n} (y_j - \bar{y})^2}. \tag{24}$$

The metrics are interpreted separately. In particular, leadership in MAE, RMSE, WAPE, or R^2 does not imply leadership in sMAPE.

3.8.2. Uncertainty metrics

Empirical coverage is:

$$\mathrm{Coverage}_{1-\alpha} = \frac{1}{n}\sum_{j=1}^{n} \mathbb{I}\big(y_j \in [\hat{L}_j, \hat{U}_j]\big). \tag{25}$$

Average interval width is:

$$\mathrm{AvgWidth}_{1-\alpha} = \frac{1}{n}\sum_{j=1}^{n} (\hat{U}_j - \hat{L}_j). \tag{26}$$

For the matched Company A conformal intervals, calibration and sharpness are summarized using the interval score:

$$\mathrm{IS}_{\alpha} = (\hat{U} - \hat{L}) + \frac{2}{\alpha}(\hat{L} - y)\mathbb{I}(y < \hat{L}) + \frac{2}{\alpha}(y - \hat{U})\mathbb{I}(y > \hat{U}). \tag{27}$$

Lower interval score is preferable. Coverage, width, and interval score are considered jointly because nominal coverage can be obtained through excessively wide intervals.

3.9. Statistical and robustness analyses

3.9.1. M5 statistical interpretation

The retained M5 outputs do not contain the raw time-indexed or series-level paired errors required for a dependence-aware comparison. Consequently, no inferential significance claim is made for M5; model differences are reported descriptively using held-out forecasting metrics.

3.9.2. Company A entity-level dependence-aware testing

Repeated daily errors from the same forecasting entity are not treated as independent. For each forecasting entity s, the MAE of CoRAL-RF is compared with the MAE of control model c:

$$W_{s,c} = \mathbb{I}\big(\mathrm{MAE}_s^{CoRAL} < \mathrm{MAE}_s^{c}\big). \tag{28}$$

Under the null hypothesis that neither model is more likely to win:

$$H_0 : P(W_{s,c} = 1) = 0.5. \tag{29}$$

A two-sided exact sign test is conducted using one paired win/loss outcome per forecasting entity. Holm adjustment is applied across the three prespecified dependence-aware comparisons:

1. CoRAL-RF versus Random Forest;
2. CoRAL-RF versus the no-Qwen gate;
3. CoRAL-RF versus the shuffled-Qwen gate.

The reported effect indicators are the proportion of forecasting entities won, the median entity-level MAE advantage, and the win-rate effect:

$$E_{win} = 2\hat{p}_{win} - 1. \tag{30}$$

The categorical-embedding comparison is reported descriptively because paired entity-level outcomes were not retained for that control. No dependence-aware significance claim is made for this comparison.

3.10. Ablation, gate, regime, and computational analyses

The contribution of the architecture is examined using dataset-appropriate ablations.

For M5, the principal variants are:

1. full CoRAL-RF;
2. CoRAL-RF without Qwen-derived context;
3. Random Forest anchor only;
4. LSTM temporal expert only;
5. BNN MC-Dropout comparator; and
6. standalone Qwen text baseline.

For Company A, the principal controls are:

1. full CoRAL-RF;
2. Random Forest;
3. LightGBM;
4. LSTM;
5. no-Qwen gate;
6. shuffled-Qwen gate; and
7. categorical-embedding gate.

The Qwen contribution is assessed separately using MAE and RMSE. A representation is not considered uniformly beneficial when one metric improves and another deteriorates.

Gate behavior is evaluated by dividing held-out observations into quartiles according to $g_{i,t}$. For each quartile, the analysis reports mean gate value, Random Forest MAE, hybrid MAE, and gain relative to the anchor:

$$\mathrm{Gain}_q = \mathrm{MAE}_q^{RF} - \mathrm{MAE}_q^{Hybrid}. \tag{31}$$

The activation rate above a prespecified diagnostic threshold is:

$$A_{0.10} = \frac{1}{n}\sum_{j=1}^{n} \mathbb{I}(g_j > 0.10). \tag{32}$$

The threshold is used only for interpretation and does not alter the prediction rule. Company A gate behavior is also evaluated separately for promotion-active and non-promotion observations and for holiday-related regimes. These diagnostics determine whether improvements are concentrated in high-gate or commercially relevant observations rather than assuming that frequent correction is desirable.

The computational evaluation reports:

1. total training time;
2. inference time;
3. peak GPU memory;
4. trainable parameter count; and
5. the incremental cost of Qwen embedding extraction relative to the RF LSTM model without Qwen.

These measures are reported because the practical value of a small forecasting gain must be considered together with the computational overhead of the contextual component.

3.11. Inventory decision evaluation

The inventory analysis is an illustrative one-period proxy rather than a complete multi-period inventory-optimization model. For model m, the planned inventory quantity is set equal to the non-negative upper endpoint of its 90% interval:

$$I_t^{(m)} = \max\left(0, \hat{U}_t^{(m)}\right). \tag{33}$$

For realized demand D_t, overstock and stockout units are:

$$\mathrm{Overstock}_t^{(m)} = \max\left(I_t^{(m)} - D_t, 0\right), \tag{34}$$

$$\mathrm{Stockout}_t^{(m)} = \max\left(D_t - I_t^{(m)}, 0\right). \tag{35}$$

The proxy cost is:

$$C^{(m)} = c_h \sum_t \mathrm{Overstock}_t^{(m)} + c_s \sum_t \mathrm{Stockout}_t^{(m)}, \tag{36}$$

where the reported experiment uses:

$$c_h = 1, \qquad c_s = 3. \tag{37}$$

Fill rate is:

$$\mathrm{FillRate}^{(m)} = 1 - \frac{\sum_t \mathrm{Stockout}_t^{(m)}}{\sum_t D_t}. \tag{38}$$

The simulation assumes zero replenishment lead time, no inventory carryover between observations, no minimum-order constraints, no capacity restrictions, and no product substitution. Therefore, the resulting costs are proxy decision indicators under the stated assumptions and are not interpreted as universal operational savings.

For Company A, the inventory evaluation compares CoRAL-RF with Random Forest, LightGBM, and the no-Qwen gate using their matched 90% conformal intervals. For M5, CoRAL-RF is compared only with Random Forest.

3.12. Non-causal price-scenario analysis

Price scenarios are evaluated only for M5 because it contains sell-price information. A counterfactual price input $p_t^{(\delta)}$ is created by applying a predefined percentage change δ while holding the other recorded predictors fixed. The model then produces:

$$\hat{D}_t^{(\delta)} = f\left(p_t^{(\delta)}, \mathbf{x}_{t,-p}\right), \tag{39}$$

where $\mathbf{x}_{t,-p}$ denotes the remaining predictors.

Scenario revenue is:

$$R_t^{(\delta)} = p_t^{(\delta)} \hat{D}_t^{(\delta)}. \tag{40}$$

When a unit-cost assumption u_t is available, scenario gross profit is:

$$\Pi_t^{(\delta)} = \left(p_t^{(\delta)} - u_t\right)\hat{D}_t^{(\delta)}. \tag{41}$$

This procedure is a predictive sensitivity analysis. It does not identify causal price elasticity because historical prices may be correlated with unobserved demand expectations, promotion decisions, inventory conditions, or managerial actions. It also does not solve a price-optimization problem.

3.13. Reproducibility and figure-generation disclosure

Reproducibility was addressed through standardized experimental settings across all evaluated models. Within each dataset, the same forecasting entities, chronological partitions, target observations, and evaluation periods were used. Model hyperparameters and gate configurations were determined using the validation partition, whereas the held-out test set was reserved exclusively for final evaluation. Fixed random seeds were used for data processing, neural-network initialization, embedding shuffling, and other stochastic procedures where applicable.

The Qwen encoder was kept frozen throughout the experiments. Its model checkpoint, tokenizer configuration, serialization scheme, pooling strategy, and embedding dimensionality were fixed and documented in the experimental configuration. For the Company A dataset, the anonymized entity identifiers, entity-selection criteria, chronological split dates, preprocessing procedure, and test indices were recorded while respecting the confidentiality constraints associated with the proprietary data.

The complete experimental workflow was implemented in Python using open-source libraries. Software dependencies, experimental

Table 6
Point-forecasting performance on the M5 held-out test set.

Model	Family	MAE	RMSE	sMAPE	WAPE (%)	R^2
CoRAL-RF	**Proposed**	**8.603**	**12.506**	39.362	**22.900**	**0.721**
Random Forest	ML	8.812	12.756	38.607	23.457	0.710
LSTM	DL	8.955	12.958	**33.877**	23.837	0.700
XGBoost	ML	9.160	12.910	37.319	24.383	0.702
PatchTST	SOTA	9.175	13.257	35.367	24.422	0.686
BNN MC-Dropout	Probabilistic	9.236	13.092	42.389	24.586	0.694
LightGBM	ML	9.461	13.691	36.220	25.182	0.665
N-HiTS	SOTA	9.513	13.718	38.280	25.323	0.664
N-BEATS	SOTA	9.749	14.105	39.627	25.951	0.645
Forecast Transformer	DL	9.910	13.992	44.186	26.378	0.651
ARIMA	Classical	10.573	14.758	42.480	28.143	0.611
TFT	SOTA	10.634	15.017	43.279	28.307	0.597
SARIMA	Classical	11.469	16.464	40.069	30.528	0.516
Qwen text baseline	Foundation model	12.320	16.020	47.870	32.793	0.542

ML: machine learning; DL: deep learning; SOTA: state-of-the-art time-series model. Metric-specific best values are shown in bold.

configurations, numerical outputs, statistical analyses, and figure-generation procedures were retained to support computational reproducibility. Quantitative figures were generated directly from the corresponding experimental outputs, whereas conceptual diagrams were constructed to represent the methodological components and their relationships as defined in the proposed framework.

4. Results

This section presents the empirical evaluation of CoRAL-RF in two retail forecasting settings. The M5 dataset is used as the principal development benchmark and supports comparisons with classical, machine-learning, deep-learning, probabilistic, advanced time-series, and foundation-model baselines. The Company A dataset is used for external validation, matched conformal comparisons, LLM-control experiments, and entity-level dependence-aware testing.

The results are organized around point-forecasting accuracy, statistical robustness, uncertainty calibration, component contribution, gate behavior, computational cost, and downstream inventory decision indicators. All configurations were selected using validation data, and the reported performance was measured on held-out test observations. Lower MAE, RMSE, sMAPE, WAPE, interval score, and inventory cost indicate better performance. Higher R^2 and fill rate are preferable, while empirical coverage is assessed relative to the nominal 90% target.

4.1. M5 point-forecasting performance

Table 6 reports the point-forecasting results on the M5 test set. CoRAL-RF achieved the lowest MAE, RMSE, and WAPE and the highest R^2 among the evaluated models, with MAE = 8.603, RMSE = 12.506, WAPE = 22.900%, and $R^2 = 0.721$.

Random Forest was the strongest standalone tabular baseline, with MAE = 8.812 and RMSE = 12.756. CoRAL-RF therefore reduced MAE by 2.37% and RMSE by 1.96% relative to Random Forest. Relative to the standalone LSTM, the reductions were 3.93% in MAE and 3.49% in RMSE. These gains indicate improved absolute-error performance, although the separation from the strongest baselines remains modest.

CoRAL-RF did not obtain the lowest sMAPE. LSTM achieved the best sMAPE of 33.877, compared with 39.362 for CoRAL-RF. Accordingly, the M5 results do not support a claim of uniform superiority across all forecasting metrics. The principal advantage of CoRAL-RF is observed under MAE, RMSE, WAPE, and R^2.

Fig. 7 summarizes the MAE rankings on M5 and Company A, while Fig. 8 presents the M5 RMSE ranking and the RMSE reductions relative to Random Forest in both datasets.

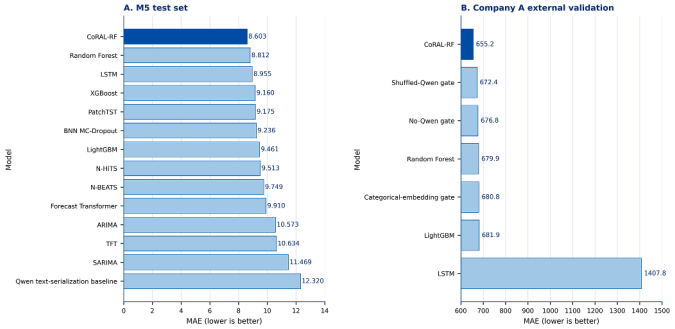


Fig. 7. MAE comparison on the M5 test set and the Company A external-validation dataset. CoRAL-RF achieved the lowest MAE in both settings, although the gains over strong tabular baselines were moderate.

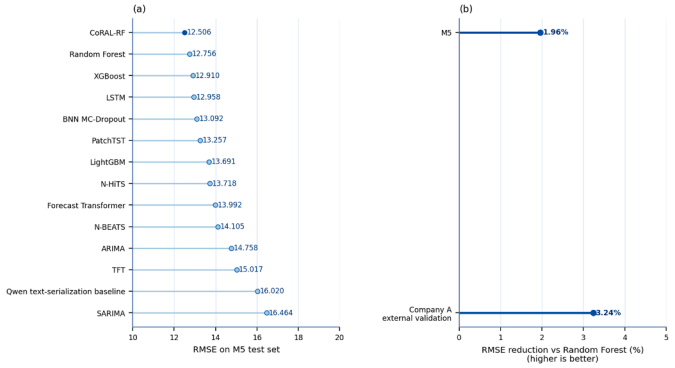


Fig. 8. RMSE results. Panel (a) presents the M5 model ranking. Panel (b) reports the RMSE reduction of CoRAL-RF relative to Random Forest on M5 and Company A.

4.2. M5 uncertainty calibration

CoRAL-RF achieved empirical 90% coverage of 0.923 on M5, with an average interval width of 41.883. The coverage is slightly above the nominal target, indicating conservative calibration on this held-out period. A CRPS value is not reported because the retained output contains only a single conformal interval and does not define the complete predictive distribution required for CRPS.

Complete matched calibration metrics for Random Forest, LightGBM, and LSTM were not retained in the aggregate M5 results. The archived Random Forest upper endpoint used in the inventory proxy

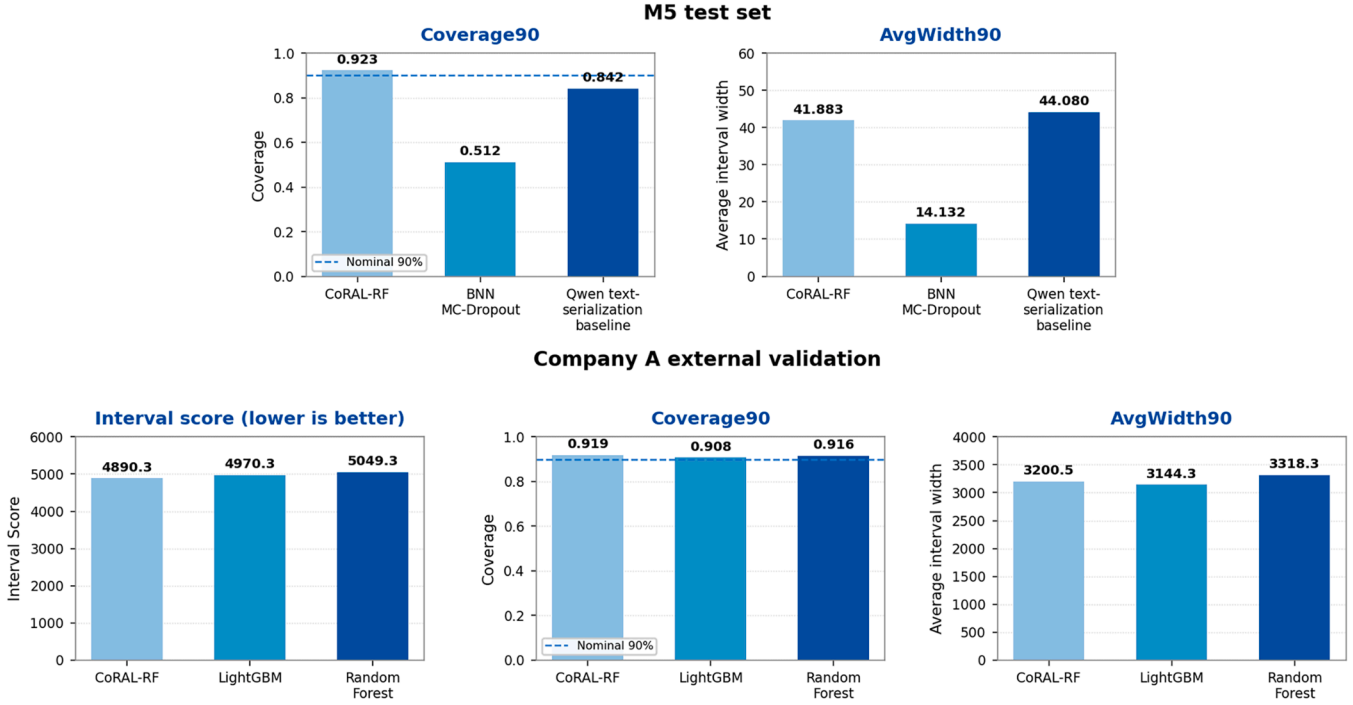


Fig. 9. Uncertainty evaluation. The M5 panels provide descriptive comparisons among different uncertainty mechanisms. The Company A panels compare models calibrated using the same nominal coverage and conformal procedure.

Table 7
M5 component ablation results.

Variant	MAE	RMSE	MAE increase vs. full (%)	RMSE increase vs. full (%)
CoRAL-RF full	**8.603**	12.506	0.000	0.000
Without Qwen context	8.669	**12.480**	0.767	-0.208
Random Forest anchor only	8.812	12.756	2.429	1.999
LSTM expert only	8.955	12.958	4.092	3.614
BNN MC-Dropout	9.236	13.092	7.358	4.686
Qwen text baseline	12.320	16.020	43.206	28.099

does not support a calibration-quality comparison. The M5 interval result should therefore not be interpreted as proving that CoRAL-RF has better calibration or sharpness than conformalized non-LLM forecasters. The fair matched comparison is instead provided in the Company A validation.

Fig. 9 combines the descriptive M5 uncertainty results with the matched Company A comparison. The M5 probabilistic alternatives use different uncertainty-generation mechanisms and are shown for context rather than as strictly matched conformal baselines.

4.3. M5 ablation and gate diagnostics

Table 7 reports the principal M5 ablations. Removing Qwen-derived context increased MAE from 8.603 to 8.669, corresponding to a 0.77% degradation. However, RMSE improved slightly from 12.506 to 12.480. The Qwen contribution is therefore mixed and metric-dependent rather than uniformly positive.

The Random Forest anchor alone obtained MAE = 8.812 and RMSE = 12.756. Relative to Random Forest, the full architecture reduced MAE by 2.37% and RMSE by 1.96%. This difference reflects the combined Random Forest–LSTM correction and contextual routing mechanism and should not be attributed to Qwen alone. The standalone Qwen text baseline was substantially weaker, confirming that Qwen is not competitive as a direct numerical forecaster in this experiment.

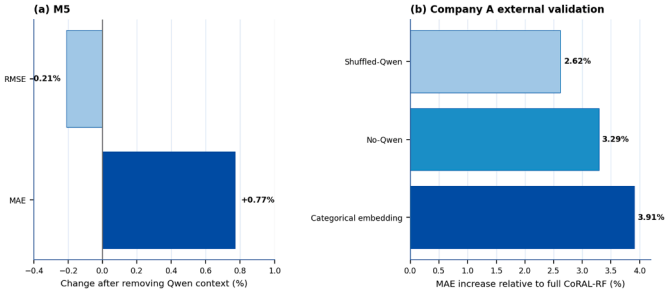


Fig. 10. Contextual representation controls. Panel (a) shows the metric-dependent effect of removing Qwen context on M5. For Company A in Panel (b), MAE increased by 3.29% for the no-Qwen gate and by 2.62% for the shuffled-Qwen gate relative to the full CoRAL-RF model.

Fig. 10 presents the Qwen-control results across both datasets. On M5, removing Qwen increased MAE by 0.77% but reduced RMSE by 0.21%. This mixed outcome supports interpreting Qwen as an auxiliary routing input rather than the main source of predictive accuracy.

The M5 gate diagnostics indicate sparse correction. The overall mean gate value was 0.0688, and 25% of test observations had gate values above 0.10. The first three gate quartiles were effectively inactive and

Table 8
M5 correction-gate diagnostics by gate quartile.

Quartile	Mean gate	RF MAE	Hybrid MAE	MAE gain
Q1	0.000	11.538	11.498	0.040
Q2	0.000	10.616	10.555	0.061
Q3	0.000	7.578	7.579	-0.001
Q4	0.275	5.517	**4.781**	**0.736**

Table 9
Point-forecasting performance on the Company A external-validation dataset.

Model	MAE	RMSE	sMAPE	WAPE (%)	R^2
CoRAL-RF	**655.234**	**1015.482**	**8.020**	**7.950**	**0.935**
Shuffled-Qwen gate	672.400	1040.200	8.400	8.200	0.929
No-Qwen gate	676.800	1045.000	8.450	8.280	0.928
Random Forest	679.868	1049.438	8.486	8.322	0.928
Categorical-embedding gate	680.827	1050.682	8.500	8.334	0.928
LightGBM	681.866	1051.324	8.543	8.346	0.928
LSTM	1407.801	1991.499	17.983	17.232	0.741

produced little or no improvement relative to Random Forest. In contrast, the fourth quartile had a mean gate value of 0.275 and reduced MAE from 5.517 to 4.781, corresponding to an absolute gain of 0.736, or approximately 13.3%.

Table 8 reports the corresponding M5 correction-gate diagnostics by gate quartile.

These results indicate that the correction mechanism does not intervene frequently. Its contribution is concentrated in a limited high-gate regime. The diagnostic is associative rather than causal and does not establish that gate activation independently causes the observed improvement.

4.4. Company A second-dataset validation

Table 9 reports the results on the Company A external-validation dataset, containing 17,793 held-out daily observations. CoRAL-RF achieved MAE = 655.234, RMSE = 1015.482, sMAPE = 8.020, WAPE = 7.950%, and $R^2 = 0.935$.

Relative to Random Forest, CoRAL-RF reduced MAE by 3.62%, RMSE by 3.24%, and WAPE by 4.47%. Relative to LightGBM, the reductions were 3.91% in MAE and 3.41% in RMSE. LSTM performed substantially worse in this entity-level setting, indicating that the tabular representation was especially important for Company A.

4.5. Company A dependence-aware statistical results

Because repeated daily observations from the same forecasting entity are dependent, the principal Company A statistical analysis uses one paired outcome per entity. Table 10 reports exact sign tests with Holm correction across the three prespecified comparisons.

CoRAL-RF outperformed Random Forest in 96 of 150 forecasting entities, corresponding to a 64.0% win rate and Holm-adjusted $p = .00228$. The median entity-level MAE advantage was 2.90%. This provides dependence-aware evidence of an improvement over Random Forest, although the median gain remains moderate.

CoRAL-RF also outperformed the no-Qwen gate in 92 forecasting entities and the shuffled-Qwen gate in 88 forecasting entities. Both comparisons remained significant after Holm adjustment. These results provide stronger evidence for a Qwen-specific and context-sensitive contribution on Company A than on M5, but the win-rate effects remain small to moderate.

The categorical-embedding comparison is descriptive because paired entity-level errors were not retained for that variant. No dependence-aware significance claim is therefore made for this control.

4.6. Company A matched conformal comparison

Table 11 compares CoRAL-RF, Random Forest, and LightGBM using the same nominal 90% coverage and matched conformal calibration procedure. CoRAL-RF achieved coverage of 0.919, average width of 3200.5, and the lowest interval score of 4890.3.

Random Forest achieved similar coverage but produced wider intervals and a higher interval score. LightGBM produced the narrowest average interval, with width 3144.3, but had slightly lower coverage and a higher interval score than CoRAL-RF. Consequently, CoRAL-RF achieved the best overall interval score, but it did not produce the narrowest intervals.

The matched Company A results demonstrate that near-nominal coverage is mainly provided by conformal calibration, while differences in interval score and width reflect the quality of the point-forecast center and residual distribution.

4.7. Company A context and gate diagnostics

The Company A control experiments provide more favorable evidence for the contextual component than the M5 ablation. Removing Qwen increased MAE by 3.29%, shuffling Qwen representations increased MAE by 2.62%, and replacing Qwen with categorical embeddings increased MAE by 3.91% relative to the full model. The categorical comparison remains descriptive because entity-level paired results are unavailable.

Despite these gains, the Company A gate was mostly inactive. Its overall mean value was 0.0151, and only 0.91% of observations exceeded 0.10. The MAE gain relative to Random Forest nevertheless increased progressively across the gate quartiles, from 8.0 in Q1 to 46.54 in Q4.

Table 12 reports the corresponding Company A correction-gate diagnostics by gate quartile.

Promotion-active observations had a slightly higher mean gate and activation rate than non-promotion observations. The activation rate above 0.10 was 1.57% during promotions and 0.38% outside promotions. The corresponding mean gains over Random Forest were 27.48 and 22.32, respectively. The correction is therefore sparse in both regimes but is somewhat more active and beneficial during promotional periods.

Fig. 11 summarizes the M5 and Company A gate-quartile patterns.

4.8. Computational cost

Table 13 reports the additional computational requirements of the Qwen-assisted configuration. On M5, the full model required 18.6 minutes of training, compared with 11.2 minutes for the RF LSTM model without Qwen. Inference time increased from 18.5 to 42.8 seconds, while peak GPU memory increased from 5.2 to 8.7 GB.

Table 10
Entity-level dependence-aware comparisons on the Company A held-out test set.

Comparison	Entities won	Win rate	Median MAE advantage (%)	Holm p	Win-rate effect
CoRAL-RF vs Random Forest	96/150	0.640	2.90	0.00228	0.280
CoRAL-RF vs No-Qwen gate	92/150	0.613	2.55	0.01370	0.227
CoRAL-RF vs Shuffled-Qwen gate	88/150	0.587	1.95	0.04087	0.173

Table 11
Matched 90% conformal uncertainty comparison on Company A.

Model	Coverage90	AvgWidth90	Interval score
CoRAL-RF	**0.919**	3200.5	**4890.3**
LightGBM	0.908	**3144.3**	4970.3
Random Forest	0.916	3318.3	5049.3

Table 12
Company A gate diagnostics by correction-gate quartile.

Quartile	Mean gate	RF MAE	Hybrid MAE	MAE gain
Q1	0.0005	1061.125	1053.125	8.000
Q2	0.0039	629.984	613.984	16.000
Q3	0.0123	521.996	493.996	28.000
Q4	0.0436	506.283	**459.742**	**46.542**

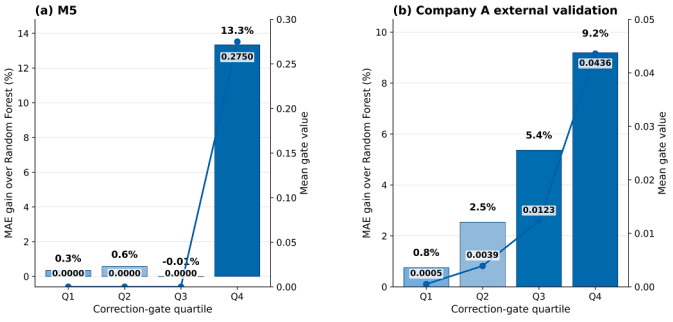


Fig. 11. Correction-gate diagnostics on M5 and Company A. Improvements are concentrated in the highest gate quartile. The gate remains inactive for most observations, particularly in the Company A validation.

On Company A, total runtime increased from 7.64 minutes without Qwen to 9.14 minutes for the full model. Inference time increased from 14.2 to 21.4 seconds, and peak memory increased from 4.9 to 7.8 GB.

These results show that the contextual component introduces non-negligible computational overhead. On M5, this additional cost accompanies only a small and metric-dependent benefit. On Company A, the stronger contextual improvement provides a clearer empirical justification, although the cost benefit trade-off remains application-dependent.

4.9. Feature-importance analysis

Feature-importance analysis was conducted using the M5 Random Forest and XGBoost models. Table 14 reports the leading combined importance scores. Rolling demand statistics and lagged sales dominated the ranking, particularly the 7-day rolling mean, lag-1 demand, 28-day rolling mean, and weekly and monthly lag variables.

These results confirm that the main predictive information originates from recent numerical demand history. They also reinforce the interpretation that Qwen-derived context is an auxiliary routing signal rather than the dominant representation in the framework.

4.10. Inventory decision results

Table 15 reports the inventory proxy results obtained by ordering at the upper endpoint of each model's 90% conformal interval, with holding cost $c_h = 1$, stockout penalty $c_s = 3$, zero lead time, and no carryover inventory.

On M5, CoRAL-RF achieved a proxy cost of 10,301.809, compared with 10,572.045 for Random Forest. This corresponds to a 2.56% reduction. The result is positive but modest and depends on the specified interval policy and cost assumptions.

On Company A, CoRAL-RF reduced proxy cost by 5.53% relative to Random Forest, from 31,386,562 to 29,650,000. However, LightGBM achieved the lowest Company A cost of 29,356,065, approximately 0.99% below CoRAL-RF. CoRAL-RF therefore does not dominate all models under the Company A inventory objective.

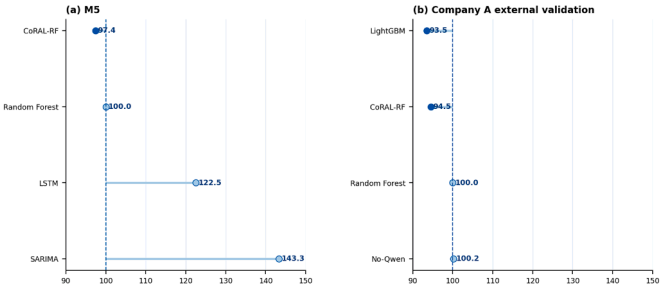


Fig. 12. Inventory proxy cost index on M5 and Company A, normalized to Random Forest = 100. Lower values indicate lower cost under the stated one-period conformal interval policy.

Fig. 12 presents normalized cost indices with Random Forest fixed at 100. CoRAL-RF achieved an index of 97.4 on M5 and 94.5 on Company A. LightGBM obtained the lowest Company A index of 93.5.

These simulations omit positive replenishment lead time, inventory carryover, minimum-order quantities, capacity constraints, substitution, and multi-period stock dynamics. The reported savings are therefore scenario-specific decision indicators rather than universal operational guarantees.

4.11. Non-causal price-scenario results

The M5 price-scenario analysis examines how model forecasts and a simple revenue indicator change when the sell-price input is varied while the remaining variables are held fixed. Table 16 reports the retained scenario outputs.

The response is non-monotonic: predicted units increase under the -10% and -5% scenarios relative to -15%, but the -5% scenario also exceeds the base prediction. Positive price scenarios produce lower predicted units than the base case and higher calculated revenue in the retained outputs. At a 15% price increase, predicted units decrease relative to the base case while calculated revenue increases by 11.287%.

These outputs are predictive sensitivity diagnostics rather than causal price-response estimates. Historical price variation may be endogenous, and the experiment does not identify price elasticity or solve a price-optimization problem.

Across the two datasets, the results consistently support a modest, dataset-dependent benefit from selective correction, while the Qwen contribution and downstream inventory ranking vary by setting. These patterns are interpreted in Section 5 without repeating the preceding numerical results.

5. Discussion

The results indicate that CoRAL-RF is a competitive framework for uncertainty-aware retail demand forecasting, but its contribution should be interpreted as incremental and dataset-dependent rather than as uniform superiority. On M5, CoRAL-RF achieved the lowest MAE, RMSE, and WAPE and the highest R^2, with MAE = 8.603, RMSE = 12.506, WAPE = 22.900%, and $R^2 = 0.721$. On the Company A external-validation dataset, it achieved MAE = 655.234, RMSE = 1015.482, WAPE = 7.950%, and $R^2 = 0.935$. These results suggest that combining a stable Random Forest anchor with a selectively applied LSTM correction can improve absolute-error forecasting across two structurally different retail settings.

Nevertheless, the improvements over the strongest tabular baselines remain moderate. On M5, CoRAL-RF improved MAE and RMSE over Random Forest by 2.37% and 1.96%, respectively, but these differences

Table 13
Computational cost of the full and no-Qwen configurations.

Dataset	Variant	Training time (min)	Inference time (s)	Peak GPU memory (GB)	Trainable parameters
M5	CoRAL-RF full	18.60	42.8	8.7	1,081,344
M5	RF LSTM without Qwen	11.20	18.5	5.2	820,000
Company A	CoRAL-RF full	9.14	21.4	7.8	1,081,344
Company A	RF LSTM without Qwen	7.64	14.2	4.9	820,000

Table 14
Top combined M5 feature-importance scores from Random Forest and XGBoost.

Feature	RF	XGB	Mean
roll_mean_7	0.361	0.213	0.287
lag_1	0.130	0.116	0.123
roll_mean_28	0.169	0.037	0.103
lag_28	0.050	0.048	0.049
lag_7	0.053	0.034	0.043
lag_14	0.041	0.042	0.042
item_id_FOODS_2_360	0.000	0.059	0.029
lag_21	0.020	0.019	0.019
dept_id_FOODS_2	0.000	0.035	0.017
day_of_week	0.005	0.009	0.007

are interpreted descriptively because dependence-aware paired errors were unavailable. On Company A, the improvements were larger and more consistently distributed across forecasting entities: CoRAL-RF reduced MAE by 3.62% and RMSE by 3.24% relative to Random Forest and achieved lower entity-level MAE in 96 of 150 forecasting entities. The cross-dataset evidence therefore supports the practical competitiveness of the framework, while also showing that the magnitude and statistical strength of the improvement depend on the dataset and inferential unit.

5.1. Interpretation of forecasting performance

The M5 results show that the Random Forest anchor already captures most of the predictable structure contained in lagged demand, rolling statistics, price, calendar, and product store variables. CoRAL-RF improves this strong baseline by applying an LSTM-derived temporal deviation only when the learned gate assigns a positive correction weight. The relatively small difference between CoRAL-RF and Random Forest is therefore consistent with the architecture: the objective is not to replace the tabular model, but to correct a limited subset of forecasts for which sequential information appears useful.

The M5 results require cautious interpretation. CoRAL-RF numerically outperformed Random Forest and LSTM, but the retained outputs do not permit dependence-aware inference. The findings are therefore interpreted as modest descriptive improvements under absolute-error metrics, not as evidence of a statistically robust separation from the strongest baselines.

The metric-specific results reinforce this conclusion. CoRAL-RF did not achieve the lowest M5 sMAPE; LSTM obtained the best relative-percentage error. This difference reflects the fact that MAE, RMSE, WAPE, and sMAPE emphasize different aspects of forecasting performance. MAE and WAPE are closely related to absolute demand deviations and inventory exposure, whereas sMAPE places more emphasis on relative error, particularly for low-volume observations. Consequently, CoRAL-RF should not be described as universally superior across all accuracy criteria. Its strongest evidence is associated with absolute-error and variance-sensitive measures.

The Company A results provide additional support for transferability. CoRAL-RF outperformed Random Forest, LightGBM, and the contextual control variants on the fixed external-validation subset. Importantly, the dependence-aware entity-level test showed that the improvement over Random Forest was not driven solely by a small number of daily observations: the full model achieved lower MAE in 64% of forecasting entities, with a median entity-level advantage of 2.90%. This represents stronger evidence than the M5 row-level comparison, although the median gain remains operationally moderate rather than transformational.

5.2. Contribution of the hybrid architecture

The ablation results support the anchor-correction design more clearly than they support any individual component. On M5, the full CoRAL-RF model outperformed the Random Forest anchor and the LSTM expert when each was used alone. Using the Random Forest anchor alone increased MAE by 2.43% and RMSE by 2.00% relative to the full model; equivalently, the full model reduced MAE by 2.37% and RMSE by 1.96% relative to Random Forest. This suggests that the principal architectural contribution arises from combining tabular stability with selective temporal correction.

The LSTM is particularly useful as a correction expert rather than as an independent forecaster. On both datasets, the standalone LSTM was weaker than the Random Forest anchor, and the difference was especially large on Company A. This indicates that the structured entity, calendar, operational, and lagged-demand features contain information that a sequence-only representation does not capture as effectively. The hybrid architecture benefits from using the LSTM to estimate deviations from a stable tabular reference rather than allowing it to determine the complete forecast.

The gate diagnostics further support this interpretation. On M5, the overall mean gate was 0.0688, and the meaningful improvement was concentrated in the highest gate quartile. In that quartile, the hybrid reduced MAE from 5.517 to 4.781 relative to the Random Forest anchor. The lower quartiles produced little improvement, and one quartile showed a negligible deterioration. On Company A, the gate was even more conservative: the mean value was 0.0151, and only 0.91% of observations exceeded 0.10. Nevertheless, the gain over Random Forest increased progressively from the first to the fourth gate quartile.

These findings show that the correction mechanism is sparse rather than continuously active. Sparse activation should not be interpreted as a failure of the gate, because the architecture is designed to preserve the Random Forest forecast when the estimated temporal deviation is unnecessary. However, the diagnostic remains associative: the concentration of gains in high-gate observations does not independently prove that gate activation causes the improvement. It shows that the learned routing scores are aligned with observations for which the hybrid correction is more useful.

5.3. Role of the Qwen-derived context

The contribution of the Qwen component differs substantially between the two datasets. On M5, removing Qwen-derived context increased MAE by only 0.77%, while RMSE improved slightly by 0.21%. This mixed result does not support the claim that Qwen uniformly improves forecasting performance. It instead suggests that the contextual representation provides a small MAE-oriented routing benefit while having negligible or slightly adverse effects on the treatment of larger errors.

The standalone Qwen text baseline performed considerably worse than the dedicated numerical models. This confirms that the language model is not an effective direct demand forecaster in the evaluated configuration. Its defensible role is therefore limited to providing an auxiliary representation for the correction gate.

Table 15
Inventory proxy results under the upper 90% conformal interval policy.

Dataset	Model	Overstock units	Stockout units	Fill rate	Total proxy cost
M5	**CoRAL-RF**	2592.079	2569.910	**0.886**	**10,301.809**
M5	Random Forest	2645.106	2642.313	0.883	10,572.045
Company A	LightGBM	25,185,546	1,390,172.875	0.9904	**29,356,064.625**
Company A	CoRAL-RF	25,900,000	1,250,000.000	**0.9922**	29,650,000.000
Company A	Random Forest	27,536,112	1,283,483.375	0.9912	31,386,562.125
Company A	No-Qwen gate	27,619,226	1,280,424.000	0.9912	31,460,498.000

Table 16
M5 non-causal price-scenario diagnostics.

Price change (%)	Predicted units	Revenue change (%)
-15	22269.533	-16.084
-10	22540.258	-10.066
-5	22802.609	-3.960
0	22554.848	0.000
5	21854.676	1.615
10	21852.248	6.450
15	21851.533	11.287

Company A provides stronger evidence for the contextual component. The no-Qwen gate had 3.29% higher MAE than the full model, while the shuffled-Qwen gate had 2.62% higher MAE. Dependence-aware comparisons also favored the full model in 92 of 150 forecasting entities relative to no-Qwen and in 88 forecasting entities relative to shuffled-Qwen, with both comparisons remaining significant after Holm correction. The shuffled control is important because it indicates that the improvement depends partly on the correct alignment between the contextual representation and the associated entity-day observation, rather than only on adding a higher-dimensional vector.

The categorical-embedding control produced weaker aggregate results than the full model, but paired entity-level errors were not retained for this variant. Consequently, the comparison remains descriptive and cannot establish that Qwen is statistically superior to conventional categorical embeddings. Overall, the evidence supports a modest and dataset-dependent Qwen contribution. The language-model representation is most appropriately described as auxiliary contextual routing information rather than the main source of predictive accuracy or a necessary component in every retail setting.

5.4. Uncertainty calibration and risk-aware forecasting

The uncertainty results demonstrate the importance of separating point forecasting from interval calibration. On M5, CoRAL-RF achieved empirical 90% coverage of 0.923 and an average interval width of 41.883. The coverage is slightly conservative on the held-out period. CRPS is not interpreted because a single retained conformal interval does not define a complete predictive distribution.

However, matched conformal versions of the strongest M5 non-LLM baselines were not available in the aggregate results. The M5 uncertainty result therefore supports the reliability of the CoRAL-RF intervals but does not establish cross-model superiority in calibration or sharpness. This distinction directly follows from the model-agnostic nature of conformal prediction: coverage validity is primarily generated by the calibration procedure, whereas the underlying point forecaster influences the interval center and residual magnitude.

The Company A analysis provides a fairer matched comparison because CoRAL-RF, Random Forest, and LightGBM were calibrated using the same nominal level and procedure. CoRAL-RF achieved the lowest interval score, 4890.3, with coverage of 0.919. Random Forest obtained similar coverage but wider intervals and a higher interval score. LightGBM produced the narrowest intervals but slightly lower coverage and a higher interval score than CoRAL-RF. Thus, CoRAL-RF achieved the most favorable combined calibration sharpness score, but it did not dominate every individual uncertainty criterion.

From a decision-support perspective, these results show why point accuracy and uncertainty quality should be evaluated jointly. A model can provide narrow intervals with inadequate coverage or reliable coverage through overly wide intervals. The matched Company A results suggest that the improved point-forecast center of CoRAL-RF can contribute to better interval scores, while the near-nominal coverage itself remains primarily a consequence of conformal calibration.

5.5. Cross-dataset robustness and generalizability

The addition of Company A addresses the limitation of evaluating the framework on only one retail benchmark. M5 contains item–store unit sales with price and hierarchical product information, whereas Company A contains anonymized daily SKU-level observed unit demand and operational predictors. Consistent absolute-error improvements across these two settings provide stronger evidence of transferability than an M5-only evaluation.

Nevertheless, the external-validation evidence remains bounded. The Company A experiment uses a fixed anonymized external-validation sample rather than the company's complete operational history, and its target scale and aggregation level differ from M5. The entity-level test improves inferential validity by avoiding the treatment of correlated daily observations as independent, but the conclusions still apply specifically to the retained forecasting entities and chronological period. Commercial restrictions prevent disclosure of the calendar boundaries and detailed field definitions, so the external experiment cannot be independently reconstructed from the paper alone.

The contrast between the two datasets is itself informative. The overall hybrid improvement and the Qwen-specific contribution were stronger on Company A, while the M5 evidence was smaller and more metric-dependent. This indicates that contextual routing may be more valuable in operational environments with heterogeneous SKU and calendar patterns than in settings where dense numerical lag and hierarchical features already explain much of the demand variation. This interpretation remains a hypothesis rather than a causal explanation and should be investigated on additional datasets.

5.6. Managerial and operational implications

The inventory analysis shows that better forecasts and calibrated intervals can translate into operational value, but the magnitude and ranking depend on the decision rule. Under the illustrative M5 policy, CoRAL-RF reduced the proxy inventory cost by 2.56% relative to Random Forest. This indicates that a modest point-forecast improvement can produce a somewhat larger cost reduction when forecast uncertainty is incorporated into the order quantity.

On Company A, CoRAL-RF reduced proxy cost by 5.53% relative to Random Forest and achieved the highest reported fill rate. However, LightGBM produced the lowest overall proxy cost, approximately 1% below CoRAL-RF. Thus, CoRAL-RF does not dominate all models under the downstream objective, even though it achieves the best point-forecast metrics. This divergence illustrates that forecast accuracy and inventory cost are related but not identical objectives. Interval width,

residual distribution, holding cost, and stockout penalties jointly determine the final policy ranking.

The simulations use the upper endpoint of the 90% conformal interval under holding cost $c_h = 1$, stockout cost $c_s = 3$, zero lead time, and no inventory carryover. These assumptions simplify the relationship between forecast uncertainty and stock decisions. Real retail systems involve positive and uncertain lead times, initial inventory, replenishment cycles, minimum order quantities, capacity constraints, supplier reliability, substitution, perishability, and multi-period stock dynamics. The reported values should therefore be interpreted as proxy decision indicators under a transparent experimental policy rather than guaranteed cost savings in operational deployment.

The practical contribution of CoRAL-RF is consequently not that it always yields the lowest inventory cost. Its value is that it provides an integrated process through which a competitive point forecast is calibrated and translated into an explicit risk-aware inventory rule. Managers could adapt the nominal coverage, cost ratio, or service target to the operational characteristics of a particular product or store.

5.7. Computational trade-offs

The contextual component introduces a measurable computational cost. On M5, the full model required 18.6 minutes of training, compared with 11.2 minutes for the RF LSTM configuration without Qwen. Inference time increased from 18.5 to 42.8 seconds, while peak GPU memory increased from 5.2 to 8.7 GB. On Company A, the full model required 9.14 minutes compared with 7.64 minutes without Qwen, and inference time and peak memory also increased.

This overhead is relevant because the Qwen benefit is small and mixed on M5. In such a setting, practitioners may reasonably prefer the simpler RF LSTM or Random Forest configuration when computational cost, latency, or deployment simplicity is more important than a small reduction in MAE. On Company A, the stronger Qwen-specific results provide a clearer justification for the additional cost, although the economic value of the gain would still depend on forecasting scale, update frequency, hardware availability, and the cost of forecast errors.

Freezing the Qwen encoder reduces training complexity and avoids repeated language-model fine-tuning, but embedding extraction remains more expensive than conventional categorical encoding. The framework should therefore be viewed as offering a configurable trade-off: the contextual block can be retained when validation evidence demonstrates sufficient incremental value and omitted when a simpler model provides a more favorable accuracy cost balance.

5.8. Role of price and promotion effects

The M5 price-scenario analysis illustrates how a demand forecast conditioned on price and promotion information can support managerial sensitivity analysis. The predicted response is non-monotonic across the full scenario range; therefore, it should be interpreted as model sensitivity rather than as an estimated demand curve. In the retained positive-price scenarios, predicted units were below the base prediction while the simple calculated revenue indicator increased.

They should not be interpreted as causal price-response estimates. Historical prices and promotions may be endogenous because they can be selected in response to expected demand, inventory conditions, competitive behavior, or managerial planning. Holding all non-price inputs fixed does not reproduce the broader market response to an actual pricing decision. The analysis reports revenue only because no validated unit-cost assumption is available for a defensible profit calculation; it also omits competitive reactions, cross-item substitution, customer retention, and long-term demand effects.

The appropriate interpretation is therefore non-causal scenario-based decision support. Establishing causal elasticity or recommending an optimal price would require an experimental or quasi-experimental identification strategy, together with an explicit optimization problem and operational constraints.

5.9. Limitations

Several limitations should be acknowledged.

First, the evaluation includes only one public benchmark and one anonymized operational dataset. Moreover, Company A represents a fixed sample rather than the company's complete operational history. Generalizability should therefore be evaluated across additional product categories, markets, aggregation levels, forecast horizons, and supply-chain settings.

Second, paired time-indexed or series-level errors were unavailable for the aggregate M5 results, preventing dependence-aware statistical comparisons. Consequently, the modest improvements over Random Forest and LSTM on M5 should be interpreted descriptively rather than inferentially.

Third, the same validation partition was used for model and gate selection and for conformal calibration. Therefore, the intervals are empirically calibrated, and the strict split-conformal coverage guarantee does not apply. Their reliability also depends on calibration residuals remaining representative of the test period. Abrupt demand shifts, disruptions, store closures, or unusual promotions may reduce coverage. Future studies should use separate validation and calibration subsets and investigate rolling, adaptive, or online conformal methods.

Fourth, the contribution of the frozen Qwen encoder is mixed: it is small and metric-dependent on M5 but more favorable on Company A. The categorical-embedding comparison is also descriptive because paired entity-level errors were unavailable. Controlled experiments should examine whether lightweight fine-tuning, smaller encoders, learned categorical representations, or distillation can provide a better balance between contextual value and computational cost.

Fifth, the correction gate is inactive for most observations, particularly on Company A. Although consistent with selective correction, this indicates that the observed gains are concentrated in a limited subset of cases. Further evaluation is required to determine whether this sparsity remains stable across periods, entities, demand regimes, and distribution shifts.

Finally, the inventory analysis is a simplified one-period proxy that assumes fixed holding and stockout costs, zero lead time, and no inventory carryover. Similarly, the price analysis is a predictive sensitivity experiment rather than a causal or optimization model. These experiments illustrate potential decision-support applications but do not reproduce the full complexity of retail operations.

6. Conclusion

This study proposed CoRAL-RF, a hybrid framework for uncertainty-aware retail demand forecasting and decision support. The model combines a Random Forest tabular anchor, an LSTM temporal correction expert, frozen Qwen-derived contextual representations, a validation-selected correction gate, and a separate conformal calibration layer. Unlike a simple output-level ensemble, CoRAL-RF preserves the Random Forest forecast as a stable reference and applies an LSTM-derived temporal correction selectively. The Qwen representation is used only as auxiliary routing information and does not directly generate the final demand forecast.

On the M5 test set, CoRAL-RF achieved the lowest MAE, RMSE, and WAPE and the highest R^2 among the evaluated models, with MAE = 8.603, RMSE = 12.506, WAPE = 22.900%, and $R^2 = 0.721$. Relative to Random Forest, the framework reduced MAE by 2.37% and RMSE by 1.96%. However, these improvements are interpreted descriptively because dependence-aware paired errors were unavailable. In addition, LSTM achieved the lowest sMAPE. The M5 findings therefore support a modest numerical advantage under absolute-error and inventory-relevant metrics rather than uniform superiority across all measures.

The Company A external validation provided stronger evidence of transferability. CoRAL-RF achieved MAE = 655.234, RMSE = 1015.482, WAPE = 7.950%, and $R^2 = 0.935$, reducing MAE and RMSE by 3.62% and 3.24%, respectively, relative to Random Forest. A dependence-aware entity-level comparison showed that CoRAL-RF outperformed Random Forest in 96 of 150 forecasting entities and remained significant after Holm correction. These results indicate that the hybrid correction mechanism can transfer across different retail structures, although the magnitude of the improvement remains moderate.

The uncertainty analysis further showed that conformal calibration produced coverage close to the nominal 90% level. On M5, CoRAL-RF achieved empirical coverage of 0.923 and average interval width of 41.883; CRPS is omitted because the retained single-interval output does not define a complete predictive distribution. Because matched conformal versions of the strongest M5 baselines were not available in the aggregate results, no claim of M5 cross-model interval superiority is made. On Company A, where matched conformal comparisons were available, CoRAL-RF achieved coverage of 0.919 and the lowest interval score of 4890.3. LightGBM produced narrower intervals, however, confirming that CoRAL-RF did not dominate every uncertainty criterion. The interval-validity benefit should therefore be attributed primarily to the conformal calibration procedure, while the point-forecast model influences interval centering, residual magnitude, and overall interval score.

The ablation results showed that the contribution of Qwen-derived context is dataset- and metric-dependent. On M5, removing Qwen slightly worsened MAE but slightly improved RMSE, providing mixed evidence for the contextual component. On Company A, the full model outperformed the no-Qwen and shuffled-Qwen controls in entity-level dependence-aware comparisons, supporting a modest context-sensitive contribution. The standalone Qwen baseline remained substantially weaker than the specialized numerical forecasters. Consequently, Qwen is best interpreted as optional auxiliary routing information rather than the dominant source of accuracy or a universally necessary component.

Gate diagnostics indicated that the temporal correction was sparse and selective. On both datasets, the largest improvements over Random Forest were concentrated in the highest gate quartile, while the gate remained inactive or close to zero for most observations. This behavior is consistent with the intended anchor correction design: the model generally preserves the Random Forest forecast and intervenes only in a limited set of cases where temporal adjustment appears useful.

From a decision-support perspective, CoRAL-RF reduced the inventory proxy cost relative to Random Forest by 2.56% on M5 and 5.53% on Company A under an illustrative policy that orders at the upper endpoint of the 90% conformal interval. However, LightGBM achieved a slightly lower Company A proxy cost than CoRAL-RF. The inventory results therefore show that improved forecasting and calibrated uncertainty can support better decisions, but they do not establish universal operational dominance. The rankings depend on interval width, holding and stockout costs, lead-time assumptions, and the selected inventory policy.

Several limitations remain. The evaluation covers one public benchmark and one operational company dataset, and the Company A analysis is based on a fixed anonymized sample rather than the company's complete operational history. Dependence-aware paired errors were not available for the M5 analysis, limiting the strength of its statistical interpretation. The Qwen component introduces additional training time, inference latency, and GPU-memory requirements, which may not be justified when its incremental benefit is small. Conformal calibration also assumes that the calibration residuals remain representative of the future test period and may require adaptive or online extensions under distribution shift. Finally, the inventory simulation assumes zero lead time, no inventory carryover, and fixed holding and stockout costs, while the price scenarios are predictive sensitivity diagnostics rather than causal price-response estimates or price-optimization results.

Future research should evaluate the framework on additional retail and supply-chain datasets, test adaptive conformal methods under non-stationarity, and compare frozen language-model embeddings with smaller encoders, categorical representations, lightweight adaptation, and model distillation. Richer decision models should incorporate uncertain lead times, supplier and capacity constraints, substitution, replenishment cycles, and multi-period inventory dynamics. Causal price and promotion analysis would additionally require an appropriate experimental or quasi-experimental identification strategy.

Overall, CoRAL-RF provides a cautious but practically relevant contribution to retail forecasting. Its principal value lies in the structured integration of a stable tabular anchor, selective temporal correction, optional contextual routing, and model-agnostic uncertainty calibration. The results support the framework as a competitive approach for risk-aware retail decision support while showing that its incremental benefits depend on the dataset, evaluation metric, computational budget, and downstream decision assumptions.

CRediT authorship contribution statement

Mohammed Zizi: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Anas Chafi:** Writing – review & editing, Validation, Supervision, Project administration; **Mohammed El Hammoumi:** Validation, Supervision.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used AI-assisted tools for language polishing, grammar checking, and formatting support. All scientific content, experimental design, analysis, interpretations, and conclusions were reviewed, verified, and approved by the authors, who take full responsibility for the content of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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